

Artificial Intelligence

Vrije Universiteit Amsterdam - Faculty of Science - Artificial Intelligence - 2025-2026

Our Artificial Intelligence (AI) Master programme is oriented towards Hybrid intelligence, where AI systems and humans collaborate. The AI master programme has a human-centric approach to AI. It is aimed at providing students with solid underpinnings in the technical and algorithmic aspects of AI, combined with a thorough understanding of human functioning from human-oriented sciences.

Students approach the study of AI both from a technical perspective, focusing on the understanding, analysis and development of novel AI algorithms, and from a societal and human perspective, looking into questions such as “how can we develop and evaluate computer-based technology that exploits knowledge about human functioning?” and “how can human and AI-technology complement each other and collaborate with each other?”

The first year includes courses that focus on core AI topics, while the second year is devoted to the specialisation “Cognitive Science”, “AI for Health" or a coherent set of electives.

During the specialization phase, students can choose to follow the regular AI Master programme to deepen and broaden the knowledge of AI techniques by building up the knowledge obtained from the compulsory courses on the core AI topics. They can choose advanced elective courses in a more specialized AI topic (e.g. Learning Machines, Knowledge Representation on the Web), specific AI application area (e.g. support of people to have a healthy lifestyle and elderly care), or relevant scientific discipline (e.g. psychology, sociology, language, biomedical sciences. Another option is to continue in the specialisation “Cognitive Science” or “AI for Health", which require an additional set of compulsory courses.

Graduates from the Master in Artificial Intelligence typically move on to pursue a career such as Data Scientists, Machine Learning Engineers, or AI specialist at leading ICT companies.

Info

Level	Master
Language	English
Duration	2 years
Type	Full Time
Credits	120 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
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Artificial Intelligence specialisation Artificial Intelligence for Health

Description

AI for Health is a specialisation in collaboration with Medical Informatics (Amsterdam UMC, UvA). In this specialisation you'll learn about Medical Informatics basics, Medical AI (with a focus on imaging techniques in medicine and natural language processing techniques in medicine), and how to combine machine learning and reasoning for health applications (VU), and AI for medical imaging (UvA/AI). Two courses will take place at Amsterdam UMC (location AMC), one at Science Park (UvA/AI), and one at the VU campus.

Programme Sections

- Artificial Intelligence specialisation Artificial Intelligence for Health compulsory courses
- Artificial Intelligence research project
- Artificial Intelligence Background constrained choice
- Artificial Intelligence Machine Learning and Intelligent Systems constrained choice
- Artificial Intelligence electives
- Artificial Intelligence compulsory courses

Artificial Intelligence specialisation Artificial Intelligence for Health compulsory courses

Courses

Course Name	Period	Credits	Code
AI for Medical Imaging	P1	6EC	XMU_0045
Medical AI	P1	6EC	XMU_0038
Machine Learning and Reasoning for Health	P2	6EC	XM_0102
Medical Informatics Basics	P5+6	6EC	XMU_0037

Artificial Intelligence research project

Courses

Course Name	Period	Credits	Code
Master Project AI	Ac. Year (September)	30EC	X_400285

Artificial Intelligence Background constrained choice

Description

Students with an Artificial Intelligence/Computer Science bachelor degree should choose the course Cognitive Psychology and its Application. Other students should choose the course Skills for AI.

Courses

Course Name	Period	Credits	Code
Cognitive Psychology and Its Applications	P1	6EC	XM_40010
Skills for AI	P1	6EC	XM_0077

Artificial Intelligence Machine Learning and Intelligent Systems constrained choice

Description

Choose at least one of the 3 courses from the available "Machine Learning and Intelligent Systems" constrained choice category.

Courses

Course Name	Period	Credits	Code
Socially Intelligent Robotics	P2	6EC	XM_0074
Data Mining Techniques	P5	6EC	X_400108
Machine Learning for the Quantified Self	P6	6EC	XM_40012

Artificial Intelligence electives

Programme Sections

- Artificial Intelligence electives Bioinformatics
- Artificial Intelligence electives Deepening AI
- Artificial Intelligence electives Health Sciences
- Artificial Intelligence electives Cognitive Science
- Artificial Intelligence other electives
- Artificial Intelligence electives Social Robotics
- Artificial Intelligence electives Language Technology

Artificial Intelligence electives Bioinformatics

Courses

Course Name	Period	Credits	Code
Fundamentals of Bioinformatics	P1	6EC	X_405052
Algorithms in Sequence Analysis	P2	6EC	X_405050
Structural Bioinformatics	P4	6EC	X_405019
Bioinformatics for Translational Medicine	P5	6EC	X_405092

Artificial Intelligence electives Deepening AI

Courses

Course Name	Period	Credits	Code
Advanced Machine Learning	P1	6EC	XM_0010
Knowledge Organization	P1	6EC	X_405065
Deep Learning	P2	6EC	XM_0083
Dynamic Programming and Reinforcement Learning	P2	6EC	XM_0093
Machine Learning and Reasoning for Health	P2	6EC	XM_0102
The Social Web	P2	6EC	X_405086
Machine Learning for Graphs	P3	6EC	XM_0119
Project Reinforcement Learning	P3	6EC	XM_0120
Learning Machines	P3, P6	6EC	XM_0061
Advanced Logic	P4	6EC	X_405048
Entrepreneurship in Analytics and AI	P4	6EC	XM_0090
Knowledge Graphs and Semantic Technologies	P4	6EC	XM_0147
Applications of Modal Logic for AI	P5	6EC	XM_0082
Data Mining Techniques	P5	6EC	X_400108
Machine Learning for the Quantified Self	P6	6EC	XM_40012

Artificial Intelligence electives Health Sciences

Courses

Course Name	Period	Credits	Code
Health Psychology	P1	6EC	AM_470730
Health Promotion and Disease Prevention	P2	6EC	AM_470811
Prevention of Mental Health Problems	P3	6EC	AM_470840

Artificial Intelligence electives Cognitive Science

Courses

Course Name	Period	Credits	Code
Cognitive Psychology and Its Applications	P1	6EC	XM_40010
Seminar Cognitive Neurosciences	P1	6EC	P_MSEMCNS_AI
Aging and age-related disorders	P2	6EC	P_MAGEDIS
Memory and Memory Disorders	P2	6EC	P_MMEMORY
Neural Models of Cognitive Processes for AI	P2	6EC	P_NEUMOD_AI
Brain Imaging	P4	6EC	P_MBRIMAG_AI

Artificial Intelligence other electives

Courses

Course Name	Period	Credits	Code
History of Digital Cultures	P3	6EC	XM_0134
Secure and Trustworthy Machine Learning	P3	6EC	XM_0179
Digital Business Ecosystems and Platforms	P4	6EC	XM_0176
ICT for the Global South	P5	6EC	XM_0142
AI for Security	P6	6EC	XM_0178
Inclusive ICT	P6	6EC	XM_0177

Artificial Intelligence electives Social Robotics

Courses

Course Name	Period	Credits	Code
Essentials of Media Psychology	P1	6EC	S_EMP
Conversational AI and Robots	P2	6EC	L_AAMPLIN041
Human-interactive Agent Learning	P4	6EC	XM_0146
Social Robotics	P4	6EC	S_SR

Artificial Intelligence electives Language Technology

Courses

Course Name	Period	Credits	Code
Subjectivity Mining	P1	6EC	L_AAMPLIN018
Conversational AI and Robots	P2	6EC	L_AAMPLIN041
Language as Data	P2	6EC	L_PAMATLW001
Machine Learning for NLP	P2	6EC	L_AAMPLIN024
Applied Text Mining 1: Methods	P3	6EC	L_PAMATLW004
Advanced NLP	P4	6EC	L_AAMPLIN035

Artificial Intelligence compulsory courses

Description

These courses are obligatory for all AI master specialisations.

Courses

Course Name	Period	Credits	Code
Evolutionary Computing	P1	6EC	X_400111
Deep Learning	P2	6EC	XM_0083
Knowledge Representation	P2	6EC	XM_0059
Multi-Agent Systems	P2	6EC	XM_0052
AI and Society	P3	6EC	XM_0075

Course Name	Period	Credits	Code
Experimental Design and Data Analysis	P4	6EC	X_405078
Natural Language Processing	P5	6EC	XM_0121

Artificial Intelligence specialisation Cognitive Science

Description

The Cognitive Science specialisation focuses on the study of human cognition through computational methods. It is jointly organised by the Department of Cognitive Psychology (Faculty of Behavioural and Movement Sciences), and the Department of Computer Science (Faculty of Sciences), and includes courses from both departments.

Students in Cognitive Science come from a wide range of backgrounds –including psychology, computer science, artificial intelligence, philosophy, mathematics, neuroscience, and others – but they all share the common goal of getting a better understanding of the human mind through computational modelling. The developed models can roughly be applied from two perspectives. Firstly, from a more theoretical perspective, cognitive models (e.g., of perception, attention, or decision making) can serve as a useful tool for researchers to gain more insight in the dynamics of cognitive processes by building (and simulating) them. Secondly, from a more practical perspective, cognitive models can serve as a basis for the development of artefacts that either show or understand human-like behaviour. Examples of artefacts that show human- like behaviour are virtual characters in (serious) games, and examples of artefacts that understand human-like behaviour are intelligent support systems in cars or in military domains.

Programme Sections

- Artificial Intelligence Background constrained choice
- Artificial Intelligence Machine Learning and Intelligent Systems constrained choice
- AI research project Cognitive Science
- Artificial Intelligence specialisation Cognitive Science compulsory courses
- Artificial Intelligence electives
- Artificial Intelligence compulsory courses

Artificial Intelligence Background constrained choice

Description

Students with an Artificial Intelligence/Computer Science bachelor degree should choose the course Cognitive Psychology and its Application. Other students should choose the course Skills for AI.

Courses

Course Name	Period	Credits	Code
Cognitive Psychology and Its Applications	P1	6EC	XM_40010
Skills for AI	P1	6EC	XM_0077

Artificial Intelligence Machine Learning and Intelligent Systems constrained choice

Description

Choose at least one of the 3 courses from the available "Machine Learning and Intelligent Systems" constrained choice category.

Courses

Course Name	Period	Credits	Code
Socially Intelligent Robotics	P2	6EC	XM_0074
Data Mining Techniques	P5	6EC	X_400108
Machine Learning for the Quantified Self	P6	6EC	XM_40012

AI research project Cognitive Science

Description

This research project (psychology department) is an option for students following the Conginitive Science specialisation.

Courses

Course Name	Period	Credits	Code
Master Project AI	Ac. Year (September)	30EC	X_400285
Master Thesis: Research Project Cognitive Science	Ac. Year (September)	30EC	P_MTHRCSC

Artificial Intelligence specialisation Cognitive Science compulsory courses

Courses

Course Name	Period	Credits	Code
Seminar Cognitive Neurosciences	P1	6EC	P_MSEMCNS_AI
Neural Models of Cognitive Processes for AI	P2	6EC	P_NEUMOD_AI
Brain Imaging	P4	6EC	P_MBRIMAG_AI

Artificial Intelligence electives

Programme Sections

- Artificial Intelligence electives Bioinformatics
- Artificial Intelligence electives Deepening AI
- Artificial Intelligence electives Health Sciences
- Artificial Intelligence electives Cognitive Science
- Artificial Intelligence other electives
- Artificial Intelligence electives Social Robotics
- Artificial Intelligence electives Language Technology

Artificial Intelligence electives Bioinformatics

Courses

Course Name	Period	Credits	Code
Fundamentals of Bioinformatics	P1	6EC	X_405052
Algorithms in Sequence Analysis	P2	6EC	X_405050
Structural Bioinformatics	P4	6EC	X_405019
Bioinformatics for Translational Medicine	P5	6EC	X_405092

Artificial Intelligence electives Deepening AI

Courses

Course Name	Period	Credits	Code
Advanced Machine Learning	P1	6EC	XM_0010
Knowledge Organization	P1	6EC	X_405065
Deep Learning	P2	6EC	XM_0083
Dynamic Programming and Reinforcement Learning	P2	6EC	XM_0093
Machine Learning and Reasoning for Health	P2	6EC	XM_0102
The Social Web	P2	6EC	X_405086
Machine Learning for Graphs	P3	6EC	XM_0119
Project Reinforcement Learning	P3	6EC	XM_0120
Learning Machines	P3, P6	6EC	XM_0061
Advanced Logic	P4	6EC	X_405048
Entrepreneurship in Analytics and AI	P4	6EC	XM_0090
Knowledge Graphs and Semantic Technologies	P4	6EC	XM_0147
Applications of Modal Logic for AI	P5	6EC	XM_0082
Data Mining Techniques	P5	6EC	X_400108

Course Name	Period	Credits	Code
Machine Learning for the Quantified Self	P6	6EC	XM_40012

Artificial Intelligence electives Health Sciences

Courses

Course Name	Period	Credits	Code
Health Psychology	P1	6EC	AM_470730
Health Promotion and Disease Prevention	P2	6EC	AM_470811
Prevention of Mental Health Problems	P3	6EC	AM_470840

Artificial Intelligence electives Cognitive Science

Courses

Course Name	Period	Credits	Code
Cognitive Psychology and Its Applications	P1	6EC	XM_40010
Seminar Cognitive Neurosciences	P1	6EC	P_MSEMCNS_AI
Aging and age-related disorders	P2	6EC	P_MAGEDIS
Memory and Memory Disorders	P2	6EC	P_MMEMORY
Neural Models of Cognitive Processes for AI	P2	6EC	P_NEUMOD_AI
Brain Imaging	P4	6EC	P_MBRIMAG_AI

Artificial Intelligence other electives

Courses

Course Name	Period	Credits	Code
History of Digital Cultures	P3	6EC	XM_0134
Secure and Trustworthy Machine Learning	P3	6EC	XM_0179
Digital Business Ecosystems and Platforms	P4	6EC	XM_0176
ICT for the Global South	P5	6EC	XM_0142
AI for Security	P6	6EC	XM_0178
Inclusive ICT	P6	6EC	XM_0177

Artificial Intelligence electives Social Robotics

Courses

Course Name	Period	Credits	Code
Essentials of Media Psychology	P1	6EC	S_EMP
Conversational AI and Robots	P2	6EC	L_AAMPLIN041
Human-interactive Agent Learning	P4	6EC	XM_0146
Social Robotics	P4	6EC	S_SR

Artificial Intelligence electives Language Technology

Courses

Course Name	Period	Credits	Code
Subjectivity Mining	P1	6EC	L_AAMPLIN018
Conversational AI and Robots	P2	6EC	L_AAMPLIN041
Language as Data	P2	6EC	L_PAMATLW001
Machine Learning for NLP	P2	6EC	L_AAMPLIN024
Applied Text Mining 1: Methods	P3	6EC	L_PAMATLW004
Advanced NLP	P4	6EC	L_AAMPLIN035

Artificial Intelligence compulsory courses

Description

These courses are obligatory for all AI master specialisations.

Courses

Course Name	Period	Credits	Code
Evolutionary Computing	P1	6EC	X_400111
Deep Learning	P2	6EC	XM_0083
Knowledge Representation	P2	6EC	XM_0059
Multi-Agent Systems	P2	6EC	XM_0052
AI and Society	P3	6EC	XM_0075
Experimental Design and Data Analysis	P4	6EC	X_405078
Natural Language Processing	P5	6EC	XM_0121

Artificial Intelligence main specialisation Artificial Intelligence

Programme Sections

- Artificial Intelligence research project
- Artificial Intelligence Background constrained choice
- Artificial Intelligence Machine Learning and Intelligent Systems constrained choice
- Artificial Intelligence electives
- Artificial Intelligence compulsory courses

Artificial Intelligence research project

Courses

Course Name	Period	Credits	Code
Master Project AI	Ac. Year (September)	30EC	X_400285

Artificial Intelligence Background constrained choice

Description

Students with an Artificial Intelligence/Computer Science bachelor degree should choose the course Cognitive Psychology and its Application. Other students should choose the course Skills for AI.

Courses

Course Name	Period	Credits	Code
Cognitive Psychology and Its Applications	P1	6EC	XM_40010
Skills for AI	P1	6EC	XM_0077

Artificial Intelligence Machine Learning and Intelligent Systems constrained choice

Description

Choose at least one of the 3 courses from the available "Machine Learning and Intelligent Systems" constrained choice category.

Courses

Course Name	Period	Credits	Code
Socially Intelligent Robotics	P2	6EC	XM_0074
Data Mining Techniques	P5	6EC	X_400108
Machine Learning for the Quantified Self	P6	6EC	XM_40012

Artificial Intelligence electives

Programme Sections

- Artificial Intelligence electives Bioinformatics
- Artificial Intelligence electives Deepening AI
- Artificial Intelligence electives Health Sciences
- Artificial Intelligence electives Cognitive Science
- Artificial Intelligence other electives
- Artificial Intelligence electives Social Robotics
- Artificial Intelligence electives Language Technology

Artificial Intelligence electives Bioinformatics

Courses

Course Name	Period	Credits	Code
Fundamentals of Bioinformatics	P1	6EC	X_405052
Algorithms in Sequence Analysis	P2	6EC	X_405050
Structural Bioinformatics	P4	6EC	X_405019
Bioinformatics for Translational Medicine	P5	6EC	X_405092

Artificial Intelligence electives Deepening AI

Courses

Course Name	Period	Credits	Code
Advanced Machine Learning	P1	6EC	XM_0010
Knowledge Organization	P1	6EC	X_405065
Deep Learning	P2	6EC	XM_0083
Dynamic Programming and Reinforcement Learning	P2	6EC	XM_0093
Machine Learning and Reasoning for Health	P2	6EC	XM_0102
The Social Web	P2	6EC	X_405086
Machine Learning for Graphs	P3	6EC	XM_0119
Project Reinforcement Learning	P3	6EC	XM_0120
Learning Machines	P3, P6	6EC	XM_0061
Advanced Logic	P4	6EC	X_405048
Entrepreneurship in Analytics and AI	P4	6EC	XM_0090
Knowledge Graphs and Semantic Technologies	P4	6EC	XM_0147
Applications of Modal Logic for AI	P5	6EC	XM_0082
Data Mining Techniques	P5	6EC	X_400108
Machine Learning for the Quantified Self	P6	6EC	XM_40012

Artificial Intelligence electives Health Sciences

Courses

Course Name	Period	Credits	Code
Health Psychology	P1	6EC	AM_470730
Health Promotion and Disease Prevention	P2	6EC	AM_470811
Prevention of Mental Health Problems	P3	6EC	AM_470840

Artificial Intelligence electives Cognitive Science

Courses

Course Name	Period	Credits	Code
Cognitive Psychology and Its Applications	P1	6EC	XM_40010
Seminar Cognitive Neurosciences	P1	6EC	P_MSEMCNS_AI
Aging and age-related disorders	P2	6EC	P_MAGEDIS

Course Name	Period	Credits	Code
Memory and Memory Disorders	P2	6EC	P_MMEMORY
Neural Models of Cognitive Processes for AI	P2	6EC	P_NEUMOD_AI
Brain Imaging	P4	6EC	P_MBRIMAG_AI

Artificial Intelligence other electives

Courses

Course Name	Period	Credits	Code
History of Digital Cultures	P3	6EC	XM_0134
Secure and Trustworthy Machine Learning	P3	6EC	XM_0179
Digital Business Ecosystems and Platforms	P4	6EC	XM_0176
ICT for the Global South	P5	6EC	XM_0142
AI for Security	P6	6EC	XM_0178
Inclusive ICT	P6	6EC	XM_0177

Artificial Intelligence electives Social Robotics

Courses

Course Name	Period	Credits	Code
Essentials of Media Psychology	P1	6EC	S_EMP
Conversational AI and Robots	P2	6EC	L_AAMPLIN041
Human-interactive Agent Learning	P4	6EC	XM_0146
Social Robotics	P4	6EC	S_SR

Artificial Intelligence electives Language Technology

Courses

Course Name	Period	Credits	Code
Subjectivity Mining	P1	6EC	L_AAMPLIN018
Conversational AI and Robots	P2	6EC	L_AAMPLIN041
Language as Data	P2	6EC	L_PAMATLW001
Machine Learning for NLP	P2	6EC	L_AAMPLIN024
Applied Text Mining 1: Methods	P3	6EC	L_PAMATLW004
Advanced NLP	P4	6EC	L_AAMPLIN035

Artificial Intelligence compulsory courses

Description

These courses are obligatory for all AI master specialisations.

Courses

Course Name	Period	Credits	Code
Evolutionary Computing	P1	6EC	X_400111
Deep Learning	P2	6EC	XM_0083
Knowledge Representation	P2	6EC	XM_0059
Multi-Agent Systems	P2	6EC	XM_0052
AI and Society	P3	6EC	XM_0075
Experimental Design and Data Analysis	P4	6EC	X_405078
Natural Language Processing	P5	6EC	XM_0121

Courses

Advanced Logic

Course Code	X_405048
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. F. van Raamsdonk
Examiner	dr. F. van Raamsdonk
Teaching Staff	dr. F. van Raamsdonk
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

The objective of the course Advanced Logic is to obtain a basic understanding of modal logic.

Knowledge and insight:

- know the Kripke semantics, and variations on that definition, for example game semantics,
- know the tableaux and sequent method for decidability,
- know the notion of bisimulation, and the game approach to (dis)proving bisimularity,
- know the nature of Hilbert style proof systems,
- know basic temporal logic and multi-model logic,
- know propositional dynamic logic and its semantics;

Applying knowledge and insight:

- be able to give proofs using Kripke semantics, game semantics,
- be able to decide on (non)validity using the tableaux or sequents method,
- be able to (dis)prove bisimilarity of two pointed models also in a game approach,
- be able to make elementary derivations in Hilbert style systems,
- be able to give proofs and show (non-)bisimilarity for basic temporal logic and multi-modal logic,
- be able to (dis)prove validity and discuss models in propositional dynamic logic;

Judgement:

- be able to (dis)prove definability of operators and frame properties in basic modal logic and also in basic temporal logic and multi-modal logic,
- be able to discuss some decidability issues;

Communication:

- be able to write elementary proofs;

Learning skills:

- have acquired some flexibility in dealing with different versions of modal logic.

Course Content

The course Advanced Logic is concerned with modal logic and its applications to computer science. We study some themes from the book Modal Logics for Open Minds by Johan van Benthem, in particular: basic modal logic and possible world semantics, bisimulation and invariance, modal definability and expressive power, decidability, proof systems, propositional dynamic logic, some epistemic logic.

Additional Information Teaching Methods

We have two lectures per week, for the duration of seven weeks. Twelve lectures discuss material and two lectures are meant for questions and wrap-up.
We have one exercise class per week, for the duration of seven weeks. We discuss (on paper) exercises.

Method of Assessment

There is a written closed-book midterm exam about the first half of the material in week 4 of the course.
There is a written closed-book final exam about all material in week 8 of the course.

If the grade for the midterm exam is higher than the grade for the final exam, then the midterm counts for 20% of the final grade (midterm-grade + 4*final-exam-grade)/5), otherwise the final grade is the grade of the final exam.

Literature

- Johan van Benthem, Modal Logics for Open Minds, CSLI Publications 2010.
- In addition we use slides and exercise sheets.

Additional Information Target Audience

MSc Artificial Intelligence
MSc Computer Science
MSc Computer Security

Explanation Canvas

We use Canvas for the schedule of the course, and to make the slides for the lectures and the exercises for the exercise classes available.
We use Canvas further to provide feedback to the homework assignments, to give the grade for the midterm, and for announcements.

Recommended background knowledge

Basic knowledge of the syntax and semantics of first-order proposition and predicate logic, as for example taught in the Bachelor course Logic and Modelling (X_401015).

Advanced Machine Learning

Course Code	XM_0010
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. S. Bhulai
Examiner	prof. dr. S. Bhulai
Teaching Staff	prof. dr. S. Bhulai
Teaching method(s)	Lecture

Course Objective

There are several learning objectives for this course. After completion of this course, the student should be able to:

- understand the capabilities and the limitations of machine learning,
- implement machine learning algorithms in Python,
- know relevant machine learning algorithms for both supervised and unsupervised learning problems,
- select the right machine learning models for real-world use cases,
- understand when to apply online learning, reinforcement learning, and deep learning,
- interpret the outcomes of machine learning algorithms.

Course Content

Machine learning is the science of getting computers to act without being explicitly programmed. It is so pervasive today that people use it in everyday life without knowing it. In this course, you will learn about the most effective machine learning techniques and gain practice implementing them and getting them to work yourself. We will discuss the theoretical underpinnings as well as the practical know-how needed to apply these techniques to new problems.

Additional Information Teaching Methods

Lectures (14 x 2 hours), including guest speakers.

Method of Assessment

Written exam.

Literature

Slides and additional material that will be posted on Canvas.

Additional Information Target Audience

mBA, mBA-D, mMath, mCS

Recommended background knowledge

The VU course Linear Algebra and the VU course Statistics, or equivalent courses, are preferred.

Advanced NLP

Course Code	L_AAMPLIN035
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Humanities
Course Coordinator	prof. dr. A.S. Fokkens
Examiner	prof. dr. A.S. Fokkens
Teaching Staff	prof. dr. A.S. Fokkens, dr. P.J.M. Sommerauer
Teaching method(s)	Seminar, Lecture

Course Objective

The goal of this course is to gain deep insight in the working of natural language processing methods. In particular, we cover:

1. how to design a system for a complex NLP task;
2. how to analyse the workings of the system through evaluation and interpretability methods;

Course Content

Natural Language Processing (NLP) is a highly dynamic research field that mainly operates on the interface between linguistics and computer science. In order to get computers to deal well with natural language, it is important to understand both how language works and how computational methods work.

This course will focus on a complex NLP task (e.g. Semantic Role Labeling, Coreference Resolution) useful in the subfield of Information Extraction. Students will be expected to carry out experiments using supervised models (including traditional, deep learning and transformer architectures). The focus of the course will be on the analysis of model behavior with the goal of identifying both strengths and weaknesses of these models.

Additional Information Teaching Methods

The course has mandatory attendance. There are two meetings per week, each lasting 1 hour and 45 mins. These meetings will be a mixture of lectures and practical sessions.

Method of Assessment

The course has three components:

- One midterm exam covering theoretical and practical concepts necessary to complete the other two components. This exam will cover: linguistic concepts relevant to the task, theoretical aspects of the machine learning models used in the course and their interaction. This component is worth 25% of the final grade and needs to be a passing grade (at least 5.5).
- One practical assignment covering the implementation of supervised learning systems for the chosen task. This component assumes prior knowledge on how to implement these systems (see ‘entry requirements’ below). This component is worth 25% of the final grade and needs to be a passing grade (at least 5.5).
- One take home exam focusing on behavioral/black-box testing of the models developed in class. This component assumes prior knowledge in core linguistic concepts in morphology, syntax and semantics. This component is worth 50% of the final grade and needs to be a passing grade (at least 5.5).

Entry Requirements

This is an advanced NLP course that assumes a solid background in linguistics, machine learning and NLP.

This course requires programming experience and the ability to independently implement and understand supervised machine learning systems (including: feature-based models such as Logistic Regression and SVMs; deep neural methods such as Bi-LSTMs; and transformer-based models such as fine-tuned BERT models).

In addition, this course also assumes prior knowledge of core linguistic concepts in morphology, syntax and semantics. Students should be able to understand concepts like syntactic ambiguity, polysemy, and the difference between words and morphemes. Students should be comfortable with constituency or dependency syntactic trees.

Students of the Language and AI/HLT programs who have successfully completed all prior courses fulfill all requirements. Other students should ensure they have had a solid course on Machine Learning (covering the range of models mentioned above). In addition, they should also have at least one or two NLP courses where the foundational linguistic concepts mentioned above were covered.

Literature

TBA

Additional Information Target Audience

Students of the Language and AI specialisation (of the Master's programme in Linguistics).
This course can also be followed by students of other Master's programmes with appropriate background knowledge.

Recommended background knowledge

Linguistics, machine learning, programming, NLP (basic concepts in these fields will be assumed to be known and not explained).

See entry requirements.

Aging and age-related disorders

Course Code	P_MAGEDIS
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. S.A.M. Sikkes
Examiner	dr. S.A.M. Sikkes
Teaching Staff	E. Butterbrod, dr. S.A.M. Sikkes
Teaching method(s)	Written partial exam, Lecture

Course Objective

The overall objective of this course is to provide advanced and up-to-date knowledge on and insight in the neuropathological, cognitive and behavioural consequences of aging and age- related neurodegenerative diseases, with a specific focus on dementia.

At the end of the course, students will be able to:

- Describe the neurobiological mechanisms of Alzheimer's disease and Related disorders, such as Lewy Body disease, Frontotemporal dementia and vascular dementia.
- Describe the societal impact related to the aging population and dementia
- Describe the difference between genetic risk, genetic susceptibility and genetic causes in the context of Alzheimer's disease, as well as discuss the most important genetic risk factors
- Explain and discuss the neurobiological trajectory from healthy aging to mild cognitive impairment and dementia, as well as potential modifying factors in this trajectory.
- Discuss whether it is possible to become extremely old without any signs of dementia from a neuropathological and functional standpoint
- Discuss the relationship between neuronal network changes and cognitive changes, and can describe the

role of network changes in the different types of dementia.
- Discuss the value of neuroimaging and biofluid biomarkers, as well as neuropsychological assessments in the context of aging and dementia.
- Critically reflect upon (non)pharmacological interventions and lifestyle prevenion in the field of dementia, and take methodological considerations into account when designing new interventions.
- Critically evaluate the concept of dementia when taking diversity into account

Course Content

This course will discuss current neurobiological, neuropsychological, and biomarker models of different neurodegenerative diseases, as well as of (extremely) healthy cognitive aging. In interactive lectures by world-leading (guest) lecturers and experts, students will learn to critically evaluate several key concepts in the field of (ab)normal aging and neurodegeneration. Recent scientific papers will be used as learning material, and students will learn how to formulate answers to research questions in a flip-the-classroom assignment.

Additional Information Teaching Methods

Plenary lectures, e-learning, assignments, flip-the-classroom assignment

Method of Assessment

The exam will consist of open-end questions, as well as a flipped-the-classroom assignment

Literature

Research papers will be used as learning material.

Additional Information Target Audience

Master students

Recommended background knowledge

The course is mainly relevant for students with a background in psychology, neuroscience, health sciences. Recommended prior reading:

Fundamental of Human Neuropsychology (Kolb & Whishaw)

AI and Society

Course Code	XM_0075
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.C.A. Klein
Examiner	dr. M.C.A. Klein
Teaching Staff	dr. M.C.A. Klein, dr. M.E.U. Ligthart, A.M. Muresan MSc
Teaching method(s)	Study Group, Lecture

Course Objective

After this course, students have a better knowledge and understanding of the societal questions that are related to AI; they are able to applying their knowledge and understanding about AI technology to contribute to the societal discussion; students are able to make judgments about societal consequences of applying AI technology.

With respect to communication skills, students have learned to articulate their informed opinions in a public debate. Students have improved their learning skill of critical thinking.

This course is particular focused on the Dublin descriptors:

- Making judgements: students have to make judgments on awareness of, and responsibility concerning, the ethical, normative and social consequences of Artificial Intelligence in the society.

- Communication skills (posters, essays, reports, and discussions).

Course Content

During this course, we will explore the role of AI in the society. Based on recent scientific literature and the portrayal of AI in contemporary movies and popular press, students and teachers will discuss topics such as the consequences of AI for the labour market and (in)equality, the ethical considerations around autonomous systems, the risks of biases and misuse of algorithms, the legal aspects of AI and the questions about the control over AI systems. In addition, we will explore possible ways to counteract negative effects. We will also reflect on the role of AI experts in this societal discussion.

Each week we discuss a specific theme, which is illustrated by a movie and presented by a guest speaker. In working group meetings, we discuss questions around the weekly theme. Attendance to the guest speaker sessions and the working group sessions is obligatory.

During the course, a number of reports based on literature have to be written, which are reviewed by other students. The final product of the course is an opinion article (essay). The course is concluded with a plenary symposium where all articles are presented in the form of a poster.

Additional Information Teaching Methods

Plenary session with movies, guest speakers, discussion meetings, group work. The attendance of guest speaker and working group sessions is obligatory.

Method of Assessment

Weekly group report + peer reviews, an individual final essay written in a controlled environment (open book exam setup), and a poster about the topic of the essay. Peer reviews are done individually and obligatory. The average of the weekly reports counts for 40%, the final essay counts for 50%, and the poster counts for 10%.

A minimum grade of 5.5 for each component is required.

There is no option for a resit.

Additional Information Target Audience

Master Artificial Intelligence (only year 1)

AI for Medical Imaging

Course Code	XMU_0045
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	S.M.A. Zaghow
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/129993

Additional Information Teaching Methods

-

Method of Assessment

-

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

AI for Security

Course Code	XM_0178
Credits	6
Period	P6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M. Zhang
Examiner	dr. M. Zhang
Teaching Staff	dr. M. Zhang, M. Chen
Teaching method(s)	Seminar, Lecture

Course Objective

This course explores a range of topics at the intersection of security and artificial intelligence (AI). The primary goal is to develop students' ability to think critically about how AI techniques can be effectively applied to address real-world security problems.

Upon completion of this course, the students will:

- Understand several security problems, e.g., software vulnerability mining, vulnerability assessment, anomaly detection (K&U, LS, C)
- Understand AI techniques, e.g., neural networks, large language models (K&U, LS)
- Have knowledge of example(s) about how to apply AI to a domain-specific problem, e.g., model security events into sequential data, connect security incidents as graphs (AK&U, MJ, LS)
- Have developed critical thinking around AI and security (MJ, LS, C)

The acronyms above refer to the Dublin Descriptors covered by each learning objective, specifically: Knowledge and Understanding (K&U), Applied Knowledge and Understanding (AK&U), Making Judgment (MJ), Learning Skills (LS), and Communication (C).

Course Content

This course explores a range of rapidly evolving security topics that have seen significant advances through the application of artificial intelligence (AI) techniques. Topics may include, but are not limited to, AI-assisted software development, automated bug discovery, automated software repair, and vulnerability assessment.

The course is structured around concrete security problems. For each topic, we begin by introducing the core security challenge, then examine how it has been traditionally addressed. We subsequently explore how AI techniques have been applied to enhance or automate solutions to the problem. Finally, we discuss the open challenges and limitations that must be addressed before AI can be effectively applied to new or emerging security scenarios.

To understand state-of-the-art methods, the course focuses on the discussion of scientific papers published at top-tier venues. Some topics may be accompanied by practical implementations; therefore, programming skills are expected.

Additional Information Teaching Methods

The course is organized into weekly lectures accompanied by practical quizzes and one final report based on a selection of five research papers. A list of papers will be provided, from which each student will choose five. The report should present a critical analysis of a specific security issue addressed in the selected papers, comparing the strengths and limitations of the approaches used. Students are expected to demonstrate critical thinking by identifying open challenges and proposing possible future directions related to the problem.

Peer Review of Reports:

Students will participate in a peer-review process by evaluating reports submitted by their classmates. Each student will provide a detailed assessment based on predefined criteria, including the quality of analysis, clarity of writing, comparison of methods, and depth of critical thinking.

Self-Evaluation of Report:

In addition to peer review, students will complete a self-evaluation of their own report. This will involve providing a detailed self-assessment against the same grading criteria, encouraging reflection on their work and the learning process.

Method of Assessment

The overall grade is determined as follows:

- Quizzes (30%)

Three quizzes, each worth 10%.

- Report (20%)

Report content: 15%, assessed based on depth of analysis, clarity, comparison of methods, and critical insight.
Peer review quality: 5%, based on the thoroughness and constructiveness of the peer reviews provided.

- Final Exam (50%)

A comprehensive exam covering the full course content.

Pass or fail
Quality of the reviews. Additional points will be deducted from students who write poor (e.g., too generic, such as "it is good") or unfair reviews.

To pass the course, you must satisfy the following criteria:

- You must score at least 6.0 on the final exam, and
- You must score 6.0 in total (i.e., with all three components taken together).

Resit
There is a resit for the final exam (50%). You will have to score 6.0 to pass the resit.
It is ***not possible*** to resit the assignments and the peer reviews.

Please note that your grade will be registered as follows:

For the first exam date: if you did not submit any quizzes, report, and did not participate in the exam, the result will be registered as no show (NS). For the second exam date: if you did not participate in the resit exam, the result will be registered as no show (NS). In all other cases, you will receive either a numeric grade or incomplete (NVD).

If you did not meet the minimum grade requirements for the exam (6.0), but your weighted average is >= 6.0, your grade is NVD. Otherwise, your final grade is your weighted average, rounded to half points except for 6.0.

If you plan to drop the course, do not submit anything, as NVD is administratively considered a failing grade.

Entry Requirements

While there are no formal entry requirements, however, students are expected to have foundational knowledge in either cybersecurity or artificial intelligence to fully benefit from the material.

Literature

No set book. All materials (i.e., papers, slides) will be made available during the course.

Additional Information Target Audience

Master Computer Security
Master Computer Science
Master Artificial Intelligence

Additional Information

Please see the information on Canvas.

Explanation Canvas

The submission of the report and the peer review of the report will be done in Canvas/FeedbackFruit.

Because of the way the system works by automatically assigning reviewers, it is not possible to grant any deadline extension for the submission of the reports or the submission of the reviews.

Students should therefore carefully plan their activities and submit also a draft version of their reports well in advance of the deadline (multiple submissions are possible) rather than waiting for the last moment.

Recommended background knowledge

While there are no formal entry requirements, however, students are expected to have foundational knowledge in either cybersecurity or artificial intelligence to fully benefit from the material. Programming skills are also expected.

Algorithms in Sequence Analysis

Course Code	X_405050
Credits	6
Period	P2
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	R. Kumar
Examiner	dr. S. Pissis
Teaching Staff	dr. J.M. Urtasun Elizari, dr. Z. Işik MSc, dr. S. Pissis, H. Verbeek MSc, prof. dr. J. Heringa, dr. ir. K.A. Feenstra
Teaching method(s)	Computer lab, Lecture

Course Objective

Have you ever wondered how we can track a gene across 3 billion years of evolution? Or how you can use the genome information of a given cancer patient to find out what may be wrong? Sequence alignment can be used to compare genomes, genes or proteins from bacteria all the way to humans, while further derived algorithms may be employed to make a phylogeny (to find out about evolutionary relationships), find a functional motif in a protein sequence, or a viral sequence in a genome. In this course we focus on the most important algorithms for biological sequence analysis that can be applied to real scientific problems in biology.

After completing the course successfully,

- You can demonstrate in-depth knowledge about the theory of sequence analysis methods.
- You know how to implement several of the most important sequence alignment and analysis algorithms.
- You understand the mathematical formulation behind sequence analysis algorithms.
- You will have understanding and skills to apply sequence analysis algorithms to protein and DNA sequences.
- You will have hands-on experience in tackling biological problems using sequence analysis algorithms, including the statistical framework of HMMs and algorithms used in genome sequencing and analysis.
- You can decide which algorithm is best suited for a particular biological sequence analysis problem.
- You can read through scientific and technical literature and learn to translate algorithms described in text and formal mathematical notation into computer code.

Course Content

Theory:

- Dynamic programming
- database searching
- pairwise and multiple alignment
- probabilistic methods including HMMs
- pattern matching
- entropy measures
- evolutionary models
- and phylogeny

Practical:

- Programming (in Python) an alignment algorithm based on dynamic programming;
- Aligning sequencing data from tumors to the human genome and analysing structural variants;
- Programming (in Python) an implementation of HMMs and using it to predict protein domain structure.

Additional Information Teaching Methods

13 lectures: 2 two-hour lectures per week.
13 computer practicals and associated assignments: 2 two-hour hands-on sessions per week.

Method of Assessment

The final grade for this course will consist of 50% practical work (see above) and 50% theoretical assessment. (assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

The theoretical assessment will be an oral and/or written exam (depending on number of students).

Further assessment and grading details will be posted on Canvas (resits and compensation rules).

Entry Requirements

Bachelor in any science discipline (including medicine).
Basic programming skills (Python) and an interest in biological problems.

Literature

Course material on Canvas.

Books: Durbin, R., Eddy, S.R., Krogh, A., Mitchison, G.. Biological Sequence Analysis. Cambridge University Press, 1998, 350 pp., ISBN 0521629713.

Recommended reading: Marketa Zvelebil and Jeremy O. Baum Understanding Bioinformatics. Garland Science 2008. ISBN-10: 0-8153-4024-9

Additional Information Target Audience

M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Artificial Intelligence
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)

We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. Ideally, this laptop supports a command line shell, Python 3 and a text editor with syntax highlighting -- either standalone (e.g. Atom or Sublime Text) or as part of a simple IDE (e.g. Spyder). As such, we recommend the Anaconda python distribution regardless of operating system, along with PuTTy or PowerShell for Windows users specifically.

The course is taught in English.

Applications of Modal Logic for AI

Course Code	XM_0082
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. V.N. Stebletsova
Examiner	dr. V.N. Stebletsova
Teaching Staff	prof. dr. K.S. Schlobach, dr. V.N. Stebletsova
Teaching method(s)	Seminar, Lecture

Course Objective

Knowledge and understanding: at the end of the course, students will be familiar with basic knowledge of some

most known Modal Logics in the field and their applications in AI. □
Applying knowledge and understanding: students will be able to formalise real-life processes based on concepts of knowledge, time, and strategic reasoning as models for appropriate logics.
Making judgements: Students will be able to reason about and assess the application decisions of Modal Logics in AI.
Communication skills: students will be able to write a scientific reports about an application of a modal logic in AI in a group of students.
Learning skills: students will be trained in writing. specifications/queries for properties of process systems based on concepts of knowledge, time, and strategic reasoning using a particular modal language and verifying those properties by model checking.

Course Content

After the first lecture which contains an introduction to the syntax and semantics of basic modal logic, we continue our lectures starting with real-life problems from various fields of AI and showing how they can be solved with different types of modal logic. Namely:

- how Epistemic Logic is useful in multi-agent field of AI, planning, decision making, in social robotics;
- how Temporal Logic is useful in planning, synthesis of provably correct strategies;
- how Dynamic Epistemic Logic is useful in epistemic planning. Epistemic planning in AI can be used for decision making in multi-agent situations with distributed knowledge and capabilities;
- how Alternating-time Temporal Logic is useful in specification and verification of open systems involving multiple autonomous agents

Our intention is to get acquainted with recent research in the field.

The course includes also practical knowledge concerning MCMAS, anopen-source, OBDD-based symbolic model checker tailored to the
verification of Multi-Agent Systems (MAS).

Additional Information Teaching Methods

The course consists of the following parts:

- lectures (two per week),
- practical seminars (once a week) which support individual practical assignments,
- working (in group of two students) on a survey,
- presentations of the survey materials.

Resit for the assignments, no resit for the group survey and presentation.

After completing lectures and individual practical assignments, each student has an oral discussion of the received knowledge with the lecturer.

Method of Assessment

Homework assignments 40% (individual work)
Survey 50% (work in groups of 2 students)
Presentation of survey 10% (in a group of 2 students, but assessed individually)

A minimum grade of 5.5 for each component is required.

Resit for the assignments, no resit for the group survey and presentation.

Literature

Reading materials — see Canvas page of the course.

Additional Information Target Audience

Master Artificial Intelligence

Recommended background knowledge

Basic knowledge of syntax and semantics of propositional and first-order predicate logic

Applied Text Mining 1: Methods

Course Code	L_PAMATLW004
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Humanities
Course Coordinator	dr. E. Maks
Examiner	dr. E. Maks
Teaching Staff	dr. L.G. de Passos Morgado da Costa BA, dr. E. Maks
Teaching method(s)	Seminar

Course Objective

In this short and intensive course, students will gain hands-on experience conducting a machine learning experiment to solve an NLP task in a group work setting.

At the end of this course students will be able to independently:

- Understand the specific needs of NLP tasks starting from literature review
- Preprocess and analyse existing NLP datasets and their annotations
- Design, set up and run machine learning experiments (developing both feature-based + transformer-based models) to solve NLP tasks
- Evaluate and interpret the results of NLP experiments
- Design and implement linguistically-informed error analyses for NLP tasks

Course Content

In this course students will work closely as a group to conduct an NLP experiment. This is a full-time course, where students are expected to commit ~40h/week and progressively work towards the completion of a series of NLP experiments. Each week, the different phases of the experiment will be covered (with deliverables).

Please note that the course is designed to be a practical course, and while some content will be reviewed, the students are expected to already have the necessary background to independently (as a group) conduct the NLP experiment.

During this course:

- Students will work on understanding the task definition and formulating an experiment design.
- Students will also work with existing datasets, understand how the data was annotated, and the need to preprocess the data and format it in a way that it can be used for the experiment;
- Students will work on building supervised classifiers for the experiment at hand. They follow standard experimental protocols.
- Students will be guided through reporting essential information in an NLP experiment, including: data preprocessing and statistics, general architecture of classifiers, features (if applicable), as well as evaluation and its metrics.
- These experimental protocols will be relevant for both traditional and transformer-based classifiers and both will be part of the course.
- Students will work on designing and implementing an error analysis for the task at hand. Students will need to understand the linguistic dimensions of the task and system’s performance. Students will analyse potential sources of error and design tests to verify their hypotheses

Additional Information Teaching Methods

This course is mostly practical. The lectures will review the basic theory related to each methodological step, which the students will apply to solve the task. A few sessions will be devoted to theoretical aspects, while other sessions will be interactive and devoted to discuss progress, doubts and problems encountered. Students will work in groups.

Students are expected to commit 40 hours per week (including both lectures and group work). Attendance is mandatory for the two weekly lectures/practical sessions.

Contact hours: 2x2 hours per week

Method of Assessment

The course includes a group+individual assignment component (with weekly deliverables) and an individual exam

component consisting of a TestVision exam and, potentially, an oral exam related to the weekly assignments. All graded components must be graded at least 5.5.

Entry Requirements

This course requires programming experience and the ability to independently implement and understand supervised machine learning systems (including: feature-based models such as Logistic Regression and SVMs; deep neural methods such as Bi-LSTMs; and transformer-based models such as fine-tuned BERT models).

In addition, this course also assumes prior knowledge of core linguistic concepts in morphology, syntax and semantics.

Students of the Language and AI/HLT programs who have successfully completed all prior courses fulfill all requirements.

Other students should ensure they have had one course on Machine Learning (e.g., Machine Learning for NLP, Machine Learning for the Quantified Self, Machine Learning). In addition, they should also have at least one NLP course where the foundational linguistic concepts mentioned above were covered (e.g., Introduction to Human Language Technology, Natural Language Processing Technology, Language as Data, Subjectivity Mining).

Literature

The literature of the course is tied to the NLP task chosen for each run of the course. Literature will mostly comprise academic papers describing a specific NLP task, and academic papers proposing systems to address that same task.

Additional Information Target Audience

The target audience of this course is mainly students in the Language and AI track of the MA in Linguistics. Other students are welcome insofar they meet the entry requirements.

Additional Information

This is a concise course requiring 40 hours of weekly commitment. Assignments are mostly group assignments and students need to contribute to these assignments equally.

Students who are not able to commit the required time should not register for this course.

Explanation Canvas

Course materials will be made available on Canvas.

Recommended background knowledge

See Entry requirements.

Bioinformatics for Translational Medicine

Course Code	X_405092
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Urtasun Elizari
Examiner	dr. E. Bosdriesz
Teaching Staff	dr. J.M. Urtasun Elizari, dr. E. Bosdriesz
Teaching method(s)	Practical, Lecture

Course Objective

Observations from biological high-throughput experiments allow us to improve diagnosis and give a personalized treatment plan for patients. However, integrating data from several sources and using this data for predictions is non-trivial.

This is a theoretical and practical Bioinformatics course on computational methods for Translational Medicine; we will focus on Bioinformatics methods that are used to predict the clinical outcome for patients and analysis methods to obtain deeper understanding of complex diseases, by combining data from various high-throughput experiments such as proteomics, microarrays and next-generation sequencing as well as existing biological databases.

After completing the course successfully,

- You can interpret research results in the context of biology
- You will be aware of Bioinformatics methods that are applicable to the area of Translational Medicine
- You can combine these methods into a creative solution using large-scale biological experiments
- You will have hands-on experience in handling large biological datasets, and will understand the complexity of the biological data both from high-throughput experiments and existing biological databases
- You will be able to work as a team to execute a large, complex biomedical bioinformatics project, and report on the results
- You will be familiar with a few in-depth research topics that lie within the expertise area of several (Bioinformatics) researchers, among others from the VU, UvA, AUMC and NKI

Course Content

Theory

- Computational analysis of molecular profiling techniques, such as WGS, WES, proteomics, RNA sequencing, and arrayCGH.
- Computational methods applied to these data types, such as: machine learning, normalisation, regularization, feature selection, classification, read mapping, clustering.
- All data analysis is relevant in a clinical setting, for diagnosis, treatment decisions or biomarker discovery.

Practical

- A large assignment for which you have to build a classifier based on molecular profiling data (the precise data-set is to be determined). You need to hand in predictions, write a paper and give a presentation. Note that this is a group project, and to participate in this practical you need to have obtained at least 12 ECT from Master-level courses.
- A smaller assignment that involves analysing and understanding molecular profiling data of another modality than the large classifier assignment.

Additional Information Teaching Methods

- 13 lectures (2 two-hour lectures per week)
- 12 computer practicals (2 two-hour sessions per week)

Method of Assessment

The final grade for this course will consist of 50% practical work (see above) and 50% theoretical assessment.

Practical assessment (50%):

- Classifier assignment (45%)
- Other data modality assignment (5%)

(assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

Theoretical assessment (50%):

- Oral or written exam (depending on number of course students)
- The exam is based on a selection of 8-10 scientific papers in the field of Bioinformatics & Translational Medicine.

Entry Requirements

Importantly: to participate in the group assignment, you need to have obtained a minimum of 12 ECT of Master level courses. In addition, some basic programming skills (either in R or Python) are required, as well as some basic knowledge on molecular biology. If you are not following the Master Bioinformatics and Systems Biology, it is advisable to first follow the Master course "Fundamentals of Bioinformatics".

Literature

- Course material on Canvas, including slides and recordings of lectures.
- 8-10 scientific papers are provided, and make up the course syllabus.

Additional Information Target Audience

M Artificial Intelligence
M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Oncology
M Computational Science

Additional Information

The course is taught in English.

- Compulsory course for students in Master Bioinformatics and Systems Biology
- Optional course for students with a Bachelor in Physics, Chemistry, Mathematics, Computer Science, Biology, or Biomedical Sciences (see requirements below).

Recommended background knowledge

If you are not following the Master Bioinformatics and Systems Biology, it is advisable to first follow the Master course "Fundamentals of Bioinformatics".

Brain Imaging

Course Code	P_MBRIMAG_AI
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. T.H.J. Knapen
Examiner	dr. T.H.J. Knapen
Teaching Staff	mr. S. Zhang MSc, mr. R. van de Klundert MSc, dr. T.H.J. Knapen
Teaching method(s)	Seminar, Lecture

Course Objective

fMRI is the main method we have of imaging activations of the living human brain, and thereby probe the computational mechanisms underpinning our cognitive repertoire. The goal of this course is to give you an idea of the awesome possibilities and annoying pitfalls of functional brain imaging, but most importantly, to give you a solid foundation for further learning. At the end of the course, you will be able to devise a valid fMRI experiment, and perform the basic analysis on the resulting data using state-of-the-art open science and open source tools.

Course Content

In this course we will teach you the ins and outs of brain imaging, that is, fMRI. We will teach you everything from the basics of signal analysis, to experimental design, to statistics. Some of the newest cutting-edge techniques, including pattern classification analysis, connectivity modelling, and resting state network analysis, are also discussed.

Additional Information Teaching Methods

Every week, there will be one or two lectures, interspersed with 3 to 4 practicals/'werkcolleges' in total. The course is broadly divided into two parts; the first half of the course serves to teach you the very basics of signal analysis and experimentation. We believe this basis is necessary to later start to think independently and academically about research in your future field. In this first phase of the course the weekly lecture will treat theory while the practicals will allow you to wet your toes with this material. This way we try to combine theory and practice.

In the second half of the course, you will already know a lot about what Brain Imaging entails. Then, we will switch gears a bit, and teach you what's going on in the neuroimaging field right now. That means that during the weekly lecture we will use research articles to illustrate the state of the art. In the practicals we'll move towards letting you perform an entire fMRI analysis yourselves. In this second part of the course we'll also focus more and more on recent articles that show us the state-of-the-art in neuroimaging.

Method of Assessment

Final Exam, open-end & MC questions 70%
Quizzes on Canvas 20%
Perusall reading participation 10%

Entry Requirements

Fluency in the Python programming language.

Literature

We will use articles describing current research.

Additional Information

Prior knowledge of Python programming and statistics is important for following the practicals.

Cognitive Psychology and Its Applications

Course Code	XM_40010
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.J. Godijn
Examiner	dr. R.J. Godijn
Teaching Staff	dr. R.J. Godijn
Teaching method(s)	Lecture

Course Objective

The course aims at providing an overview of cognitive psychology and its applications. Students will:

- understand the role of human cognitive capabilities and limitations in the design of products, work places, and large systems. (Knowledge and understanding)
- get acquainted with main areas of human factors and with main theories and findings on human performance. (Knowledge and understanding)
- understand where in the process to apply knowledge and what methods can be used to analyze human performance. (Apply knowledge and understanding)
- learn how to approach and solve an applied problem in man-machine systems. (Make judgements) (Lifelong learning skills)

Course Content

The course covers a number of central principles from the area of cognitive psychology and how these principles can be applied in the design of modern man-machine systems including human-computer systems. A variety of topics will be discussed such as the basics of cognitive psychology, research methods, attention and multitasking, mental workload, decision-making, display design, environmental design, human error and its causes, automation, and various applications such as driving behavior.

Additional Information Teaching Methods

12 Lectures

Method of Assessment

Multiple choice exam
resit multiple choice exam

Literature

Stone, Chapparro, Keebler, Chapparro, & McConnell (2025). "Introduction to human factors: Applying psychology to design" (2nd edition).

Additional Information Target Audience

The course Cognitive Psychology and its Applications is for students from AI, Psychology or any other interdisciplinary field.

Additional Information

The course Cognitive Psychology and its Applications is a core course in the MSc Artificial Intelligence. Two core qualifications of a graduate are: -"Has basic knowledge of physiological, psychological, or social aspects of human functioning that can be exploited in Socially Aware Artificially Intelligent systems." -"Has in-depth knowledge, insight and skills in at least one area of the human-oriented disciplines, e.g. Clinical and Cognitive Psychology, Social Sciences, Movement Sciences, Criminology, or Medicine." The course Cognitive Psychology and its Applications contributes to these two qualifications.

Conversational AI and Robots

Course Code	L_AAMPLIN041
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Humanities
Course Coordinator	prof. dr. P.T.J.M. Vossen
Examiner	prof. dr. P.T.J.M. Vossen
Teaching Staff	prof. dr. P.T.J.M. Vossen
Teaching method(s)	Seminar

Course Objective

Did you ever wonder how to talk to a robot in your own language? We introduce you to our robot to experience how it is.

Robots are fun to work with. Especially social robots that can communicate with people. Robots are also very complex laboratories in which you can you can implement multimodal communication models and study how people communicate with robots and vice versa. In this course, you will experiment with our robot model Leolani and learn about the complexity of multimodal communication in real-world physical contexts. You will learn about the design, the complex components and skills and the underlying data structures.

You will also learn how to evaluate conversational agents.

Course Content

You will learn about the architecture of a communicating robot and its different components, starting from the sensors that pick up signals (audio and vision), the language understanding and generation models and the brain in which the robot stores all interpretations of the encounters with people. You will read overview papers that explain the state-of-the-art of human-robot communication and you will interact with the robot and report on the findings. You will gain a deep insight in what it takes to build a communicative robot, what the difficulties are and how to evaluate multimodal interactions.

In the Spinoza research project we study the relation between the world and how people and robots make reference to the world. The central theme of this course is the relation between identity, reference and perspective. What things do we identify in a shared space or situation, how do we refer to these things and what is the perspective of the communicating partners when referring in different ways.

All students will interact with the robot and evaluate the interactions in different ways. You will write an individual report on the interaction and the evaluations.

In addition, you work in small teams on a student project related to our research.

Additional Information Teaching Methods

There are two classes per week, each two hours. There will be regular classes in which we discuss literature, explain the robot models and explain the major concepts and there will be hands on sessions during which you work in small groups with the robot.

Method of Assessment

You will write a report on the group project and an individual report on your interactions with the robot and the evaluation carried out.

Additional Information Target Audience

Students Research Master Humanities, Track Linguistics - Human Language Technology; students Master Artificial Intelligence

Recommended background knowledge

Students do not need to program but if they want they can. The programming language is Python. Students without programming skills can do experiments on the multimodal communication and the impact on people.

Data Mining Techniques

Course Code	X_400108
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M. Hoogendoorn
Examiner	prof. dr. M. Hoogendoorn
Teaching Staff	dr. S. Yu, prof. dr. M. Hoogendoorn, B. Borsboom
Teaching method(s)	Computer lab, Lecture

Course Objective

The aim of the course is that students acquire data mining knowledge and skills that they can apply in a business environment. More precisely, the following learning goals are distinguished:

1. Understand the data mining process
2. Understand exploratory data analysis
3. Understand feature engineering
4. Understand classification techniques
5. Understand regression techniques
6. Understand association rules and recommender systems
7. Understand big data engineering and big data infrastructures
8. Understand current state of the art in data mining
9. Understand ethical concerns and evaluation of results
10. Learning to argue for choices (rationale)

The course if focused on the Dublin descriptors Applying Knowledge and Understanding (since the aim is to improve the practical skills); Making judgements (which technique is appropriate, how to best apply Data Mining for a specific case); Communication skills (how to report on your approach, choices and your results), and Learning skills (able to find new relevant techniques, assess their suitability, etc.).

How the aims are to be achieved: Students will acquire knowledge and skills mainly through the following: an overview of the most common data mining algorithms and techniques (in lectures), a survey of typical and interesting data mining applications, and practical assignments to gain "hands on" experience. The application of skills in a business environment will be simulated through various assignments of the course.

Course Content

The course is intended to introduce Data Mining Techniques to students that are new to the field as well as to more experienced students. The main aim is to gain a more practical perspective towards Data Mining Techniques/Machine Learning. Lectures will cover more basic things for those new to the field (general introduction into Data Mining, classical algorithms such as decision trees, association rules, neural networks, ensemble learning, etc.) and on top will discuss advanced topics including deep learning, recommender systems, big data infrastructures, and text mining. A number of successful applications in the area will also be discussed. In addition to lectures, there will be an extensive practical part, where students will experiment with various data mining algorithms and data sets. The grade for the course will be based on these practical assignments.

Additional Information Teaching Methods

Lectures (h) and practical sessions (pra). Lectures are planned to be interactive: there will be small questions, etc.

Method of Assessment

Practical assignments. There will be two assignments done in groups of three. For the first assignment there is a choice: going for a basic version of the assignment (suited for those new to the domain) or a more advanced one (for students with more experience). The second assignment is the same for all and will involve an in-class Kaggle competition. The first assignment counts for 40% and the second for 60%.

An interview will take place on an individual student level to make sure that (1) each student understands the content of the graded material, and (2) has made a sufficient contribution to the group work. Only when this check is passed will a final grade be given. Furthermore, the grade of both assignments needs to be sufficient to pass the course. Finally, attendance of the final lecture (in which the final ranking of the Kaggle competition is announced and selected groups of students will be asked to present their solution) is mandatory as it contributes to the learning objectives of the course.

There is an option to resit one of the assignments when the assignment has been submitted but an insufficient grade is obtained (only when the other assignment grade is sufficient). This involves improving your assignment to at least a sufficient level within two weeks after having received your grade for the second assignment. The maximum grade of the resit is a 6.0.

Literature

Optional: Ian H. Witten, Eibe Frank, Mark A. Hall, and Christopher J. Pal. Data Mining: Practical Machine Learning Tools and Techniques (Fourth Edition). Morgan Kaufmann, 2016 978-0128042915.

Additional Information Target Audience

Master Artificial Intelligence, Master Computer Science, Master Bioinformatics and Systems Biology, Master Business Analytics

Recommended background knowledge

1. Statistics
2. Programming Skills

Deep Learning

Course Code	XM_0083
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S. Yu
Examiner	dr. S. Yu
Teaching Staff	dr. S. Yu, dr. P. Bloem, dr. M. Cochez
Teaching method(s)	Practical, Lecture

Course Objective

1. Knowledge and understanding

- Be able to explain the main components of a deep learning architecture: fully-connected layers, convolutional layers, activation functions, LSTM, and GRU.
- Be able to explain and derive the backpropagation algorithm.
- Be able to explain different commonly used deep architectures.
- Be able to explain Variational Autoencoders and unsupervised representation learning.
- Be able to explain the difference between discriminative and generative models.

2. Applying knowledge and understanding

- How before-mentioned layers work and where to use them.

- How to formulate a neural network for a specific problem.
- How to analyze the performance of a neural network.
- What neural network fits best for a given problem.

3. Making judgments

- What deep learning model to use for a given problem (e.g., generative vs discriminative).
- What layers of a neural network are suitable for a given problem.

4. Communication skills

- Presenting an analysis in a written form (a short report) for each assignment.

5. Learning skills

- Able to read (some) state-of-the-art papers.
- Able to develop (to some extent) available deep learning libraries.

Course Content

Deep learning becomes the leading learning and modeling paradigm in machine learning. During this course, we will present basic components of deep learning, such as:

- different layers (e.g., linear layers, convolutional layers, pooling layers, recurrent layers);
- non-linear activation functions (e.g., sigmoid, ReLU);
- backpropagation;
- learning algorithms (e.g., ADAM);
- other (e.g., dropout).

Further, we will show how to build deep architectures like AlexNet and ResNet. We will explain potential pitfalls and possible solutions, e.g., by using residual connections and dense architectures.

After discussing discriminative models, we will turn them into generative models. We will start with linear latent variable models like the probabilistic PCA. Then we will discuss a non-linear version of the pPCA, namely, Variational Auto-Encoders (VAEs). Both pPCA and VAE are so-called prescribed models that require formulating the likelihood function. Next, we will introduce unsupervised representation learning, different ways to improve VAEs, and how to obtain disentangled representations.

At the end of the course, we will outline recent developments in deep learning. Namely, we will discuss Reinforcement Learning and Deep Reinforcement Learning. In the end, we will introduce generalization and explainabilility - two fundamental topics of deep learning.

Additional Information Teaching Methods

The course consists of two parts: a digital exam and practical assignments. The digital exam is supported by lectures (two/three per week). The assignments are supported by practical sessions led by TAs.

The first two assignments are carried out individually and the remaining assignments are done in small groups. No resit is possible for the practical assignments.

Method of Assessment

Final exam (50%) and practical assignments (50%). There are 4 assignments (2 of them are done individually; 2 of them are done in groups).

The final exam must be passed with a sufficient grade (equivalent to a grade of 5.5 or higher).

There is a resit for the exam. The practical assignments cannot be redone.

Literature

The literature will be made available on Canvas.
Two suggested books:

- Goodfellow, I., Bengio, Y., & Courville, A. (2016). "Deep learning". MIT press.
- Tomczak, J. M. (2022). "Deep Generative Modeling". Springer, Cham.

Additional Information Target Audience

Master Artificial Intelligence

Master Business Analytics

Recommended background knowledge

- Calculus
- Linear Algebra
- Statistics & Probability Theory
- Programming (Python)
- Machine Learning

Digital Business Ecosystems and Platforms

Course Code	XM_0176
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J. Gordijn
Examiner	dr. J. Gordijn
Teaching Staff	dr. J. Gordijn
Teaching method(s)	Seminar, Lecture

Course Objective

An ecosystem is a collection of parties who work cooperatively and competitively to satisfy customer needs. Well-known examples are Google, Facebook, and Netflix, but there are many other eco-systems in the field, e.g. in Fintech, energy, and automotive industry. In this course, we focus on digital ecosystems which are enabled by Internet- and related technologies. Frequently, these ecosystems are disrupted by innovative information technologies, such as machine learning, blockchain, and 5G mobile. The overall goal of this course is to analyze disruptive opportunities for an eco-system given a particular emerging technology. This year, that technology is blockchain.

Upon completion of this course, students will:

- have gained knowledge and understanding of the concept of ‘eco-system’ (Knowledge and understanding);
- have gained knowledge and understanding of various methods for designing and analyzing eco-systems and their e-services, including e3value, trust modelling, BPMN, and UML class modelling (Knowledge and understanding);
- have gained knowledge and understanding of the selected technology of this year in order to analyze, design and implement a disruptive eco-system (Knowledge and understanding);
- be able to analyze and design an eco-system offering an e-service from different viewpoints (Applying knowledge and understanding);
- be able to elicit different e-service (business) ideas that can be provided in an eco-system (Applying knowledge and understanding);
- be able to explore the e-service idea from a business perspective. In this way, the student will be able to further design and develop the e-service idea using different analysis techniques (Applying knowledge and understanding);
- be able to transform the business perspective of the e-service design into a design reflecting the IT perspective (Applying knowledge and understanding);
- be able to implement (part of) the e-service using the selected disruptive technology (Applying knowledge and understanding);
- be able to make informed decisions about eco-systems and their e-services based on possible risks and opportunities (Making judgments);
- be able to assess the quality and correctness of the various conceptual models created (Making judgments);
- be able to pitch a new idea for an eco-system/e-service to investors (Communication skills);
- be able to concisely present the selected solution (Communication skills);
- be able to write a scientifically well-founded report about the eco-system/e-service under development in a group of students (Communication skills);
- have learned to work in a project group and keep each other accountable for agreed results (Learning skills);
- have learned to improve themselves based on various feedback mechanisms (Learning skills).

Course Content

The course comprises the following topics:

- Eco-systems and platforms
- Introduction in the selected technologyof this year
- Business models and value proposition design
- The e3value method for understanding business networks
- The Business Process Modelling Notation for analysing cross-organizational networks
- The Unified Modelling Language – class modelling
- Transaction modelling
- Technological design issues

Additional Information Teaching Methods

The following methods are used:

- Lectures twice a week. These lectures include in-class workshops to practice the various methods used in this course
- Group work: teams of about 2/3 students design an eco-system. At the beginning of the course a number of eco- systems are offered as inspiration
- Group feedback sessions: Each week, students will get personal feedback in face-to-face feedback sessions. Moreover, groups will receive feedback on their submitted reports by email.

Method of Assessment

Individual exam on the theory (50%), Teamwork project assignments (50%).

To pass the course, the scores of each component must be 5.5. or higher.
Compensation: it is not possible to compensate one component with another.
Mode of re-examination: if insufficient, each component can be revised and resubmitted once.

Literature

A (selection of a) list of papers on the course topics. Material distributed on Canvas.

Additional Information Target Audience

Master Artificial Intelligence
Master Computer Science
Master Information Sciences

Custom Course Registration

Further information for this course will be made available online (Canvas).
All students must be enrolled in the course Canvas community. Group enrolment takes place in Canvas.

Recommended background knowledge

The student should have sufficient knowledge about Requirements Engineering and Conceptual Modelling and should master such a modelling method (at least UML class or activity diagrams).

Dynamic Programming and Reinforcement Learning

Course Code	XM_0093
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. G.M. Koole
Examiner	prof. dr. G.M. Koole
Teaching Staff	V.G.M. Francois Lavet, prof. dr. G.M. Koole
Teaching method(s)	Seminar, Lecture

Course Objective

Dynamic Programming (DP) and Reinforcement Learning (RL) are fields concerned with decision making over time. After completing this course, the student

- is familiar with the commonly used algorithms for solving dynamic optimization problems;
- understands the main features of these algorithms, their strengths and weaknesses including their convergence properties;
- can implement them in an appropriate language;
- can model real-world decision problems into a DP or RL framework and solve moderately sized problems;
- has knowledge of the historical development of DP and RL and has an idea of possible future developments.

Course Content

This course is concerned with reinforcement learning and its origin dynamic programming. These are fields dealing with goal-directed decision making over time, such as finding your way in an unknown area, playing a game or pricing airline tickets.
We look at these areas from different angles:

- we deal with full-information "planning" problems, but also with partial-information "learning" problems
- we consider different algorithms, some of which are guaranteed to find the best solution, but also heuristics
- we consider high-dimensional problems (such as games) and methods to solve them
- we look at small toy problems to understand algorithms and sharpen our intuition, but also bigger problems for which we learn how to implement algorithms (in python)
- we look at different types of applications, both from AI (search problems, games) and OR

Additional Information Teaching Methods

Lectures and practical work integrated

Method of Assessment

Programming exercises and final exam.
The 3 assignments each count for 10% and the exam for 70%.
The minimal passing grade for the exam is 5.0.

Literature

Slides and lecture notes

Additional Information Target Audience

mBA, mAI, mCS, mBa-D, mMath

Recommended background knowledge

Programming experience in python

Entrepreneurship in Analytics and AI

Course Code	XM_0090
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W. Stam
Examiner	prof. dr. W. Stam
Teaching Staff	drs. G.J. Beekman MA, P. Afonso Branco Ramos Pinto, prof. dr. W. Stam, H.V. Nguyen
Teaching method(s)	Seminar, Written partial exam, Lecture

Course Objective

Academic and research skills: Students are able to generate a promising new business idea in the field of AI & Analytics and develop a prototype that validates the proposed customer needs and market potential.

Bridging theory and practice: Students understand the core theories and concepts in the field of entrepreneurship, as reflected by their ability to analyze, compare, and apply these theories and concepts to analyze and solve the key management challenges faced by real-world entrepreneurs.

Entrepreneurial soft skills: Students are able to effectively work in a team for generating and developing a new venture idea and can pitch the business concept to external stakeholders in a professional, engaging manner.

Course Content

Entrepreneurship in Analytics and AI is an engaging course for students who like to learn about entrepreneurship, its role and importance in our society, and the process by which entrepreneurs transform new ideas into successful AI-driven business ventures.

In this course, students will learn the ropes of this emerging field via three educational pillars:

1. Students will learn to understand and apply theories from economics, sociology, and psychology to study key topics in entrepreneurship, including innovation, new venture creation, entrepreneurial competences, entrepreneurial finance, scaling and growth of new ventures, corporate entrepreneurship, and entrepreneurial ecosystems. Theoretical understanding of these subjects is applied to real world cases focusing on key management challenges faced by entrepreneurs and how to creatively overcome these challenges.
2. Students will familiarize with a set of entrepreneurial soft skills, which they will put into practice when they will be asked to approach different stakeholders to further their team projects to gain commitments or to obtain important information. For this purpose, the students will receive training in generating initial ideas for opportunities as well as networking and network utilization.
3. Students will be introduced to the actions and strategies that promote the process of new venture creation. For this purpose, the students, supported by the lecturers and coaching, will write (and pitch) a Student Design Sprint report (including prototype) around an innovative new venture opportunity in the field of Analytics and/or AI that they deem interesting for entrance into the market.

Additional Information Teaching Methods

During this course, students will attend both (guest) lectures and tutorials. The lectures explain relevant concepts, theories, and frameworks in the field of entrepreneurship and illustrate their usefulness using various practical examples. The tutorials are designed to facilitate students in making progress with their team projects by providing appropriate guidance and feedback.

Method of Assessment

The final grade for this course is based on a written individual exam, several individual pass/fail assignments, and a Student Design Sprint (SDS) project assignment.

The written exam and pass/fail assignments are individual assessments whereas the SDS assignment is a team assessment. To pass the course, the exam and SDS assignment should be graded as sufficient. Both forms of assessment have one option for re-sit in case students received an insufficient score.

Literature

The literature for this course constitutes a selection of academic articles and case studies.

Essentials of Media Psychology

Course Code	S_EMP
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Social Sciences
Course Coordinator	dr. D.A. Parry
Examiner	dr. D.A. Parry
Teaching Staff	dr. C.C. Cabañas González, dr. K.E. Balint, dr. D.A. Parry, prof. dr. E.A. Konijn, prof. dr. T. Hartmann, dr. D.F. Preciado Vanegas, A.M.M. Leroux MSc
Teaching method(s)	Study Group, Lecture

Course Objective

Upon completion of this course, the student is able to:

- Articulate the core themes in the field of Media Psychology.
- Describe and evaluate the main theories of Media Psychology.

- Discuss and compare key findings in fundamental areas of Media Psychology such as selective media use, message processing, and media effects related to, for example, learning from media; media violence; media reality; moral issues, media, and child development; the role of emotions in media processing and reality perceptions; virtual worlds; video gaming; and the future of new media and communication technology (e.g., social robots).
- Apply a critical-reflective attitude about up-to-date and interdisciplinary scientific research in the field in an area of their choice.
- Understand and apply an academic attitude, writing style, and search for insights in complex phenomena.
- Write a proper paper introduction, formulate relevant and testable hypotheses, reflect on methodological choices and research designs, and analyze and interpret research data.

Course Content

Media Psychology is on the rise as a relatively new research field, although a number of research efforts bridging psychology and media effects research have existed for some time. Media Psychology today focuses on individual differences and psychological mechanisms underlying media effects and media use in general, thereby including the selection of particular content, differences in perceptions and experiences that underlie differences in effects. This course includes both traditional mass media (TV, film), and increasingly focuses on newer media (social media, video games, VR, social robotics), on the one hand discussing a continuous line of theorizing and on the other hand discussing differences and new approaches. This unique course will focus on the essentials of Media Psychology, discuss core lines of research, compare key findings in fundamental areas of Media Psychology and critically reflect on its implications. Key areas are: Selective media use, message processing and learning from media, media violence, media addiction, the role of emotions in media processing, media-based reality, psycho(physio)logical responses to media, morality issues in media, interactive media and digital gaming, media and psychological well-being, and the future of new media and communication technology, also including social robots.

Within these lines of research, questions that will be addressed are, among others: What is the need for a distinct field of media psychology? Why is there a debate on media violence? How do media affect individuals' brains, bodies, and behaviors, and who is most susceptible to these effects? How do we know the difference between fiction/fake and reality as represented in media fare? How can new media and communication technology contribute to the well-being of individuals and society at large? What media psychological theories can be built on when designing and adapting new technologies, and how can we use them to harness our fascination with media in positive ways?

Additional Information Teaching Methods

Lectures and workgroups.

Method of Assessment

Individual digital examination and short, individual- and group-based graded assignments.

Literature

A selection of chapters from recent books and a selection of specific journal articles will be available via online databases.

Additional Information Target Audience

- Master students in Communication Science, track Media Psychology.
- Master students AI, HRI, HCI, Psychology, Pedagogy (Pediatrics), Education, and others are welcome.

Additional Information

This is an English-language course (including assessments and assignments). Foreign exchange students are especially welcomed to join this class. This course is obligatory for master students who are following the track Media Psychology at FSW-VU.

Evolutionary Computing

Course Code	X_400111
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English

Faculty	Faculty of Science
Course Coordinator	prof. dr. A.E. Eiben
Examiner	prof. dr. A.E. Eiben
Teaching Staff	dr. K. da Silva Miras de Araujo, prof. dr. A.E. Eiben, mr. J.M. di Matteo
Teaching method(s)	Seminar, Lecture

Course Objective

This course has a threefold objective:

1. (Knowledge & Understanding) To learn about computational methods based on Darwinian principles of evolution.
2. (Applying knowledge and understanding, Communication) To use such methods as problem solvers and as simulation tools; to determine adequate algorithm setups depending on the problem at hand.
3. (Applying knowledge and understanding, Lifelong learning skills) To gain hands-on experience in performing computational experiments with evolutionary algorithms.

Course Content

This course is about constructing, applying and studying algorithms based on the Darwinian evolution theory. Driven by selection (survival of the fittest, mating of the fittest) and randomised reproduction (mutation, recombination), an evolutionary process is being emulated and solutions for a given problem are being "bred". During this course, various flavours within evolutionary computing are treated, including genetic algorithms, evolution strategies, evolutionary programming, genetic programming, differential evolution, particle swarm optimisation. Applications in optimisation, constraint handling, machine learning, and robotics are discussed. Specific subjects handled include: genetic structures (representations), selection techniques, sexual and asexual reproduction operators, (self-)adaptivity and methodological aspects, such as algorithm design & tuning, and performance assessment. Special attention is paid to the field of evolutionary robotics where the bodies (morphology, hardware) and the brains (controller, software) of robots are evolved simultaneously. Hands-on-experience with evolutionary algorithms is gained through a compulsory programming assignment.

Additional Information Teaching Methods

Oral lectures and compulsory Python programming assignment (in teams of 3 to 5). A limited number of highly motivated students can replace the programming assignment by a special research assignment under the personal supervision of the lecturer(s). These research projects aim at publications with the students as first authors of the paper.

Method of Assessment

Written exam and programming assignment (in teams of 3 to 5). To pass the course as a whole, you must pass both the exam and the programming assignment. For the final grade they will count with the same weight:

Grade = 0.5 x Exam_Grade + 0.5 x Assignment_Grade

Note that:

- The exam is multiple choice, taken on the campus in an electronic form. There is a resit in case you cannot attend or fail.
- The programming assignment is graded based on your code, the quality of the solution your algorithm finds, and a short report. **No resit is possible for the programming assignment.**
- If you pass one component (Exam or Assignment), but fail the other, then you can keep the grade of the passed component for next year or later. Then you can pass the course by getting a good grade (> 5.5) for the missing component and using the old grade in the averaging formula.

Literature

The course is based on the textbook ***A.E. Eiben and J.E. Smith, Introduction to Evolutionary Computing, Springer, 2015, 2nd edition, ISBN 978-3-662-44873-1***. The lectures and the book chapters are synchronised to a great extent. In addition, a number of carefully selected papers to provide extra information on specific topics are available via Canvas.

Additional Information Target Audience

MSc Artificial Intelligence
MSc Econometrics and Operations Research
MSc Finance

MSc Business Analytics
MSc Computer Science
MSc Computer Security
MSc Bioinformatics and Systems Biology

Additional Information

Attending the lectures in real life is not mandatory, but highly recommended. Active participation in the team work for the programming assignment is a must!

Explanation Canvas

The complete schedule of the course, organised by the chapters of the textbook, will be made available at the beginning of the course. Students are assumed to consult the course schedule to keep up with the progress, as Canvas rubrics are not used. Video recordings of the lectures made during the COVID period are still available on Canvas for self-study.

Recommended background knowledge

Python programming skills are necessary to do the practical assignment. If you cannot program, you cannot complete this course.

Experimental Design and Data Analysis

Course Code	X_405078
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E.N. Belitser
Examiner	dr. E.N. Belitser
Teaching Staff	dr. E.N. Belitser
Teaching method(s)	Practical, Lecture

Course Objective

In this course, students will become familiar with the most commonly used experimental designs, regression models, some nonparametric tests and bootstrap methods. By the end of the course, students will be able to:

- Identify and describe some of the most widely used statistical models,
- Understand the basic theory underlying these models,
- Apply these models appropriately in relevant contexts.

Course Content

The EDDA course introduces a range of widely used statistical models and techniques, including paired and independent t-tests, non-parametric rank-based tests, bootstrap and permutation methods, analysis of variance (ANOVA), ANCOVA, lasso, linear regression models, generalized linear and nonlinear models, and survival analysis. Students will learn how to choose appropriate models for various scenarios, estimate model parameters using available data, and validate model fit. For regression models, proper study design is essential to draw sound conclusions; the course covers well-known experimental designs such as completely randomized and randomized block designs, along with their associated ANOVA techniques. A portion of the course focuses on non-parametric and resampling methods, such as the Wilcoxon test (one- and two-sample), Kolmogorov–Smirnov test, rank correlation tests, and both permutation and bootstrap tests. The methods will be illustrated using practical examples, with an emphasis on real datasets. The statistical software R will be used for data analysis throughout the course.

Additional Information Teaching Methods

Lectures, practical sessions.

Method of Assessment

Course assessment consists of assignments (33.3%) carried out in teamwork during the course and an individual

digital test (66.7%) at the end. To pass the course, both components must meet minimum required thresholds. Re-examination policy: if the final result is insufficient, only the individual test may be retaken.

Literature

Literature will be communicated on Canvas.

Additional Information Target Audience

MSc AI, CS

Additional Information

Assignments are to be made using the statistical package R (http://www.r-project.org).

Explanation Canvas

All the information about the course will be available on Canvas.

Recommended background knowledge

- A basic course in statistics of the same level as Statistical Methods (X_401020) or Statistical Methods for AI (XB_0080) is mandatory. All the material and its prerequisites (basic mathematical and programming skills) for these courses will be assumed known in the lectures/assignments/exam of the course EDDA.
- A good background in probability, at least of the same level as in Statistical Methods (X_401020) or Statistical Methods for AI (XB_0080).
- Basic knowledge of the statistical software R and its application to data analysis.

Fundamentals of Bioinformatics

Course Code	X_405052
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Urtasun Elizari
Examiner	dr. ir. K.A. Feenstra
Teaching Staff	dr. J.M. Urtasun Elizari, dr. ir. K.A. Feenstra, dr. H. Mouhib, dr. D. Molenaar
Teaching method(s)	Seminar, Computer lab, Written partial exam, Lecture

Course Objective

Are you interested in Bioinformatics? Do you want to find out how biology can make an exciting application domain? Or do you want to learn what more you could do with your data, and with less effort? Enter here to start!

Fundamentals of Bioinformatics (FoB) is the starting course of the Bioinformatics master. It aims to give you a broad overview of important topics relevant to the field, with a focus on current open problems. You will be made aware of these open problems during practical sessions that will cover active research in a major Bioinformatics problem (mutation impact prediction). Based on your background, you will be assigned to separate classes where you will be working to fill gaps in your background knowledge in programming and/or biology.

After completing the course successfully:

- You can explain the fundamental principles of cell biology and the genetic information flow
- You can write short code to solve simple problems and complete skeleton scripts for biological applications
- You can read algebra and calculus equations
- You can become aware of gaps in your own background knowledge.
- You can address gaps in your background knowledge (programming, mathematics and molecular biology) and skills
- You are aware of the major issues when developing and benchmarking new methodology in bioinformatics research
- You can present and communicate a joined group projects carried out with students of diverse backgrounds (Interdisciplinarity) in a respectful way
- You can apply and understand fundamental methods in bioinformatics, e.g. sequence alignment, impact

prediction, and to understand and use fundamental concepts in bioinformatics, e.g. homology, evolutionary conservation, and gene function

Fundamental knowledge:

Students will have the opportunity to level up their knowledge of molecular biology, maths (linear algebra and calculus) and programming (Python), according to their needs.

Course Content

Topics:

- Evolution, genomics, sequence alignment, Blast/PSI-Blast, FAIR data, benchmarking, machine learning, genome assembly, mutation impact prediction, protein stucture analysis, substitution matrices, scientific writing

Project:

- Group project to benchmark different mutation impact prediction approaches. The groups will be composed to include different BSc backgrounds.

Background knowledge (1 out of 3, see below):

- Molecular cell biology
- Programming and computational thinking in Python
- Calculus and linear algebra

Additional Information Teaching Methods

- 8 lectures (two-hour lecture in the morning, two days per week) Exercises during/in between lectures
- 12 computer practicals for the group project - compulsory attendance (online or on campus) (two hour sessions following the morning lectures, two days per week), supervised.
- Feedback (theoretical and practical) will be given during the computer practical sessions.

Conversion classes for background knowledge:

- Molecular cell biology (7 x double lectures)
- Programming and computational thinking in Python (10 computer practicals)

Depending on your background, you will be assigned 1 of the 2 conversion classes. Students that are already well versed in both topics (molecular biology and programming) may be assigned to the following class, which is organised as part of the Introduction to Systems Biology:

- Calculus and linear algebra (10 maths classes)

Method of Assessment

- Project work [~40%] :
- Project report and/or presentation
- Digital exam to assess [~60%]:
- Lecture topics
- Concept learned during the project
- Concepts from conversion classes

(assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

Further assessment and grading details will be posted on Canvas (resits and compensation rules)

Entry Requirements

Bachelor in any science discipline (including medicine) or strong programming background, and an interest in biological problems.

Literature

- course material (slides, scientific papers) on canvas.vu.nl
- book: Marketa Zvelebil and Jeremy O. Baum Understanding Bioinformatics Garland Science 2008 ISBN-10: 0-8153-4024-9

Additional Information Target Audience

M Artificial Intelligence
M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Oncology
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)
We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. We will help you to set this up.

Recommended background knowledge

Programming in Python, Molecular Biology, some basics of calculus and linear algebra

Health Promotion and Disease Prevention

Course Code	AM_470811
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.C. Adriaanse
Examiner	dr. M.C. Adriaanse
Teaching Staff	E.L. Ibouanga MSc, dr. M.C. Adriaanse
Teaching method(s)	Study Group, Lecture

Course Objective

General aims

In this course, the students will learn to:

1. Provide a solid basis in understanding elementary aspects of evidence- and theory-based development and planning of health promotion and disease prevention programs.
2. Write a scientific study protocol about the planned development and evaluation of a preventive health intervention.

Learning objectives

After the course, students:

- Are able to describe and apply the elementary aspects of evidence- and theory-based development and planning of health promotion and disease prevention programs using an intervention mapping approach
- Are able to apply theoretical methods that can help to change important determinants of the environment and or health behaviors (lifestyle factors)
- Have knowledge about and is able to apply strategies that provides opportunities and challenges for the development and implementation of preventive health interventions, such as:

- Environment, habits and self-regulation
- Community action
- Interventions in public space
- Evaluating effects

- Are able to describe, understand and discuss the fundamental aspects of quality of life such as measurements, types and criteria
- Are able to describe the principals and perspectives of implementation and process-, effect and economic evaluation of health promotion programs
- Are able to integrate the acquired knowledge of the course in writing a study protocol

Course Content

This course fits in the program of the specialization Prevention and Public Health. Within this specialization you are trained to become a health promotor who is able to work in a theory- & evidence-based way and is able to link research, practice and policy. The courses within this specialization are structured according to the six steps of Intervention Mapping. These steps are:

1. Needs assessment;
2. Preparing matrices of change objectives;
3. Selecting theory-informed intervention methods and practical applications;
4. Producing program components and materials;
5. Planning program adoption, implementation, and sustainability;
6. Planning for evaluation.

The course Health Promotion and Disease Prevention will introduce you to the six steps of Intervention Mapping. Specific emphasize will be put on step 2 and 3 with a focus on primary prevention.

This course focuses on lifestyle/ health behaviors and environmental differences related to health and diseases among individuals and populations. The ultimate goal is to improve peoples' health status and quality of life by health promotion interventions.
Some examples of the topics that will be addressed are:

- Intervention mapping; designing theory- and evidence-based health promotion programs.
- Theory-based intervention methods and strategies; theoretical methods that can help to change several of the most important determinants of health behaviors.
- Environmental influences on health. The physical environment and health interact. The importance of environmental interventions and their effect on health are postulated.
- Health-related quality of life; the role of perceived mental and physical health status in the development of interventions.
- Effect and process evaluation; principals, perspectives on process evaluation, and determining the effects of health promotion programs.

Core element in this course is writing a study protocol in English, describing the design of a health promoting or disease preventing intervention trial.

Additional Information Teaching Methods

This course is rewarded with 6 ECTs and runs in Nov-Dec 2025. Health Promotion and Disease Prevention is a part-time course, this means that 21 hours a week are necessary to pursuit the goals of this course.

Teaching activities include: Lectures, tutorials, guest lecturers, group assignment (study protocol), peer review sessions and self-study.

Method of Assessment

Grades will be based on the assignment (study protocol) and a written exam that includes multiple choice and open-ended questions. The final grade is being determined by the study protocol (30%) and written exam (70%). The study protocol as well as the written exam must have a grade 5.5 or higher.

Entry Requirements

At the start of this course, we expect you to master knowledge, insight, attitude and skills at a level which is comparable to the final qualifications stated by the Bachelor Health Sciences at the VU.

Literature

The following book is used by students who follow the specialization Prevention and Public Health. Bartholomew, Eldredge, Parcel, Kok, Gottlieb, Fernandez (eds) Planning health promotion programs: an intervention mapping approach; fourth edition(2016). Jossey-Bass, San Fransisco. ISBN:978-1-119-03549-7. Chapters which are applicable to this course will be announced through Canvas. In addition, students will use a course manual, and additional course materials are provided via Canvas.

Additional Information Target Audience

The course ‘Health promotion and disease prevention’ is a compulsory course within the Master Health Science specialization Prevention and public health, and an elective course for students of other specializations of the master Health Sciences. Students with a Bachelor degree or pre-masters in Health Sciences with interest in the field of prevention and public health.

Additional Information

This course will be offered for the final time within the MSc Health Sciences in the 2025–2026 academic year.

Custom Course Registration

Registration for this course via the registration module on your VU Dashboard. Registration for the assignment in subgroups and the workgroups via Canvas.

Recommended background knowledge

Students are expected to comprehend the knowledge of the VU bachelor health science courses ‘Preventie’ and ‘Gezondheidscommunicatie’, or the VU premaster health science course ‘Preventie en volksgezondheid’ or similar courses on other universities.

Health Psychology

Course Code	AM_470730
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. B.M. van Dongen
Examiner	dr. B.M. van Dongen
Teaching Staff	dr. B.M. van Dongen, E.L. Ibouanga MSc
Teaching method(s)	Study Group, Lecture

Course Objective

- Students can explain what the field of health psychology is about in the context of tertiary prevention and which psychological aspects are relevant in the treatment of diseases and coping with chronic diseases.
- Students can recognize and explain the fundamental elements, underlying theory and evidence-based of the following central psychological aspects: hospitalization and stressful medical procedures, coping, self-management, adherence, social support, doctor-patient communication, health literacy and relapse.
- Students can explain the relationships and differences between these aspects and can provide suggestions on how to influence these aspects in an evidence-based manner in the context of tertiary prevention.
- Students are able to recognize aspects in the reality of patients and can reflect on these aspects if and in what way real-life experiences are in accordance with the theory of the chosen concept.
- Students are able to develop a feasible and scientifically supported mHealth (mobile/app) intervention plan aimed at improving self-management, coping and/or adherence

Course Content

This course fits in the program of the specialization Prevention and Public Health. Within this specialization you are trained to become a health promoter who is able to work in a theory- & evidence-based way and is able to link research, practice and policy.

Health Psychology refers to the psychological aspects of health, illness and the health care system. In the current course ‘Health Psychology’, key psychological aspects that are relevant in the treatment of diseases and coping with chronic diseases will be studied and discussed, as well as the way we can influence these aspects. As such, this course focusses on tertiary prevention, e.g. the prevention of worsening of (chronic) diseases and improving self-reliance of patients. Questions to be studied will be for example ‘How can we improve adherence of patients with diabetes with regard to their medication?’, and ‘How can we improve communication between health care workers and their patients?’. These and other questions will be studied in several cases about key psychological aspects, which will be applied to real-life practice of patients (individual) and intervention development (group) through practical assignments. The first step will be to study underlying determinants or psychological processes of the (health) problems. Second, interventions to tackle the presented problems or research into the different problems will be studied.

Additional Information Teaching Methods

This course is rewarded with 6 ECTs. Health Psychology is a part-time course, this means that 21 hours a week are necessary to pursuit the goals of this course. Attendance is mandatory for all patient-guest speakers.

During the course ‘Health Psychology’ you will obtain knowledge and understanding of:

1. Hospitalization and stressful medical procedures;
2. Coping with diseases;
3. Self-management interventions in chronic illness;
4. Adherence to advice and treatment
5. Communication processes between health care workers and their patients;
6. Health literacy and its influence on coping/ self-management
7. The role of social support
8. Relapse in health-related behavior among patients

The course is built around these topics, which will be discussed and studied during seminars. In preparation for these seminars, literature and knowledge clips need to be studied. You will use the information from the seminars to fulfill two assignments: The objective of the first assignment is to get insight into the link between theory and practice. You will link theory to personal experiences of patients who suffer from a chronic disease to reflect on lived experiences in relation to the theory. The objective of the second assignment is to learn how to develop the outlines of an evidence-based (mHealth) intervention idea in the context of tertiary prevention using Intervention Mapping.

Method of Assessment

In order to pass for the course, you must:

1. Attend all guest lectures by patients (pass/fail);
2. Pass an individually written report reflecting on key psychological aspects based applied to the experiences of the guest lecturers and the theory around these aspects (pass mark is 5.5 or higher);
3. Write a scientific report for the systematic development of an M-health Intervention. You will carry out this assignment in groups (pass mark is 5.5 or higher);

The final mark for the course is determined by:

- Assignment 1 (50%) – an individually written report based on guest lectures by patients;
- Assignment 2 (50%) – a combined grade based on the scientific report the intervention plan and a peer-assessment score.

Information regarding the resit of the assignments will be announced later through Canvas.

Entry Requirements

At the start of this course, we expect you to master knowledge, insight, attitude and skills at a level which is comparable to the final qualifications stated by the Bachelor Health Sciences at the VU.

Knowledge about theories and models with respect to psycho-social determinants of behavior is necessary, including knowledge on differences between health, health behavior and determinants of behavior and familiarity with central behavioral determinants.

Literature

The following book is required for students who follow the specialization Prevention and Public Health:

- Bartholomew Eldredge, Parcel, Kok, Gottlieb, Fernandez (eds). Planning health promotion programs: an intervention mapping approach; fourth edition (2016). Jossey-Bass, San Fransisco. ISBN: 978-1-119-03549-7 (E-book also available)

Other literature, such as book chapters and articles, will be announced on Canvas.

Additional Information Target Audience

Master students Health Sciences. All other students need approval of the course coordinator and the examination committee of their own program.

This course will be taught in in English, with the exception of one guest lecture by a patient.

Additional Information

This course will be offered for the final time within the MSc Health Sciences in the 2025–2026 academic year.

Custom Course Registration

Registration for this course via VU-net*.

*Note: final division in workgroups will be done in the first week of the course and teachers reserve the right to

make this division based on group work. Registration for workgroups in VU-net will NOT be leading.

History of Digital Cultures

Course Code	XM_0134
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D.J. Beckers
Examiner	dr. D.J. Beckers
Teaching Staff	dr. D.J. Beckers
Teaching method(s)	Seminar, Lecture

Course Objective

Students acquire a firm historical knowledge of computing in its cultural context. They will recognize the various strands in theories of “the information society”, of “digital culture”, and of "algorithmic thinking", and are aware of the strengths and limitations of such theories. (Knowledge and understanding) (Applying knowledge and understanding) (Making judgements)

Students gain experience in teamwork in a project. (Communication) (Lifelong learning skills)

Through the experience of thinking along the lines of a different discipline (i.c. history), students of computer science get the habit of making their own discipline the subject of reflection in a systematic way. (Making judgements) (Communication) (Lifelong learning skills)

Course Content

History of Digital Cultures is a hands-on reconnaissance tour through the history of the information society, from computers and software to the internet, and the present ubiquity of digital cultures.

The computer may be considered an information machine, an automatic calculator, or control technology. Three traditions build up the background for the advent of the stored program computer around 1950: data processing, process control, and scientific computation. The advent of the modern computer is treated here in its cultural context. An historical view on software is a major topic in this course. With the shaping of the internet, computers have sunk into the background. Computing and algorithms have not.

Treating the world as "data" has become second nature. History will help us recognise these aspects of today's culture. Throughout history, speculations on the societal impact of computing have accompanied this technology. History of Digital Cultures addresses computing in society and on a more abstract level theories of “the information society”, of “digital culture”, and of "algorithmic thinking"

History is concrete and local. The projects, building up the course, will be concerted efforts to explore local or national developments, possibly concerning The Netherlands, like Dutch computer- and software pioneers, contributions to software engineering, and more recent specialities like the Dutch role in the internet, "De Digitale Stad" (DDS), web design, gaming, and digital art.

The course is an intensive, engaging, and fully committing four-week project in teamwork. We expect students to be working full-time on the project. To actively participate in their own team (on site), but also reacting to the presentations of the other teams during the weekly presentations. The teaching team will be available "24/7" to assist teams and provide feedback.

Teams of 4 students choose their topic; the choice is open but we present options (with literature) on the Canvas site which you may use for inspiration.

Additional Information Teaching Methods

Seminars / working independently on project / feedback meetings / lectures / excursion.

Each lecture (except the first one) will finish with a short individual reflection on what the lecture contributed to the own research project. Every Friday there is a seminar where students present their progress to the teaching team and their fellow students. By providing and receiving (peer) feedback during these meetings, we also get inspiration from each other. Attendance of the Friday seminars is mandatory. During the weeks, students will be working independently on their group projects, but several feedback meetings are scheduled. During these meetings,

groups report on their projects to TAs, receive feedback and may pose questions. One excursion at least is scheduled, participation is on a voluntary basis.

Method of Assessment

Project assignments, presentations, project report.

The weekly presentations and draft reports are graded tentatively. Most important is the feedback the teaching team provides.

The final grade is calculated on the basis of:

1. Individual reflections to the lectures (pass/fail; two fails are allowed)
2. End presentation by the group (20%)
3. End report (50%)
4. Individual oral exam on the content of literature, lectures and project (30%) must be passed with 5.5 minimum

Literature

We start by reading and collectively discussing David Gugerli's "How the world got into the computer". More material will be announced on Canvas.

Additional Information Target Audience

MSc Computer Science

Human-interactive Agent Learning

Course Code	XM_0146
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. K. Baraka
Examiner	dr. ir. K. Baraka
Teaching Staff	dr. K.S. Luck, dr. ir. K. Baraka
Teaching method(s)	Study Group, Lecture

Course Objective

At the end of the course, students will be:

- familiar with basic knowledge of some of the core aspects of human-interactive agent learning (HIAL), including imitation learning, learning from different kinds of human feedback, preference-based learning, teaching signals, interaction problems in HIAL, and evaluation metrics. (Knowledge and understanding)
- able to program, evaluate, and discuss results of basic HIAL algorithms, as well as critically design an interaction between a human teacher and a learning robot. (Applying knowledge and understanding, Making judgements)
- able to engage in groupwork and orally present results of an open-ended research-flavored project. (Communication skills)
- trained in acquiring a set of interdisciplinary HIAL-related topics at the intersection of machine learning, human factors, and robotics, making them equipped for engaging in research in this emerging field (Learning skills)

Course Content

Human-Interactive Agent Learning (HIAL) is an area of interactive AI that focuses on developing agents (e.g., robots, artificial assistants) that can improve their task performance through interaction with humans. This course aims to cover basic HIAL principles and techniques, drawing from the fields of Machine Learning, Human Factors, Behavior Modelling, Robotics, and Design.

The course will delve into different classes of interactive machine learning algorithms that can be used to build HIAL systems, including imitation learning (e.g., behavioral cloning, inverse reinforcement learning), learning from evaluative feedback (e.g., reward shaping), preference-based learning (e.g., learning from ranked trajectories), and learning from corrective feedback (e.g, Expert Intervention Learning). In addition to learning algorithms, students

will be exposed to human-centered concepts related to situated interaction with such systems, including teaching signals, usability, human bias and limitations, transparency, alignment and synergy.

The course is meant to provide students with a good algorithmic foundation while fostering broader critical thinking about human-agent interactions.

Additional Information Teaching Methods

The course will combine lectures with individual in-lab practical assignments. There will be a small group assignment towards the end of the course, as well as a final exam.

There will be a significant amount of self-study from lecture materials and readings.

Method of Assessment

Digital exam (50%) at least 5.5/10 to pass the course
Individual practicals (30%)
Group project (20%)

There will be a resit exam, but NO resit for the group project.

Entry Requirements

This course is open to Master AI students. Any other student will need special permission to follow the course.

Literature

Academic papers and other literature will be shared during the course to complement lecture content.

Additional Information Target Audience

MSc Artificial Intelligence (elective)

Recommended background knowledge

Basic concepts of Reinforcement Learning
Basic concepts of Probability Theory and Statistics
Basic concepts of Calculus

ICT for the Global South

Course Code	XM_0142
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K. Aslam
Examiner	dr. K. Aslam
Teaching Staff	dr. J. Gordijn, dr. K. Aslam
Teaching method(s)	Seminar, Lecture

Course Objective

At the end of the course, students will be able to:

- Gain a comprehensive understanding of the digital divide, challenges and opportunities in the Global South and the role of ICT in addressing them. (Knowledge and Understanding)
- Apply technical skills and innovative approaches to develop innovative ICT solutions to meet specific needs in the Global South. (Applying Knowledge and Understanding)
- Evaluate the impact of ICT policy and governance on socio-economic development and how it contributes(d) to digital divide by considering ethical, cultural, and sustainability factors. (Making Judgements)
- Effectively communicate project design and outcomes both orally and in academic reports. (Communication)
- Develop the capacity for self-directed learning about emerging technologies and their potential applications in ICT for the Global South. (Lifelong Learning Skills)

Course Content

This course explores the role of Information and Communication Technologies (ICTs) in addressing socio-economic challenges faced by developing countries, commonly referred to as the Global South. It provides a comprehensive understanding of how ICTs can be harnessed to foster development, improve governance, enhance education, healthcare, and empower communities.

The lectures (in the form of interactive sessions) will explore how technological innovations can meet specific local needs through sustainable and culturally relevant solutions. Students will engage in hands-on projects to apply their technical skills and innovative thinking to develop real-world ICT solutions tailored to the nuanced demands of the Global South.

Students will be guided during the course to develop a reflective practice that includes empathy and ethical considerations, essential for developing ICT solutions for low resource environments. The course focuses that ICTs are powerful enablers for achieving many of the United Nations Sustainable Development Goals (SDGs), teaching students how to leverage technology to address some of the world's most pressing challenges.

Additional Information Teaching Methods

This lectures will be delivered in an interactive seminar-style format. Prior to each class, students are expected to engage with assigned readings and multimedia resources to build a foundational knowledge base. During the lecture, students will discuss main insights and reflect on their own viewpoint in relevance to the pre-class materials.

Method of Assessment

During the course students will work in a group of 3-5 (depending on the total number of students) to derive a solution for a particular problem in the context of Global South. The projects will be research-based and along the way students will need to submit four assignments in total:

Assignment #1: group (a written report of overall description of project stating goal and research questions) - 15%
Assignment#2: group (a written report of detailed description of methodology and plan to conduct the study) - 15%
Assignment#3: group - (a written report of results achieved, discussion of results in the specific context for which study is conducted and future outlook of the project) 40%
Assignment#4: individual oral presentations - 20%
Attendance: 10%

Each of the assignments should be at least graded with a 5.5 or higher and the total marks should be 6 or higher for a student to pass the course. One resubmission is allowed for Assignment#1 and Assignment#2 in case of achieved grade below 5.5. There is no resubmission for Assignment#3 and Assignment#4.

There will be no final exam.

Literature

Relevant literature will be made available on Canvas.

Additional Information Target Audience

MSc AI, MSc CS, MSc IS

Recommended background knowledge

We expect students to have at least a basic knowledge of programming and how ICT projects are conducted. Familiarity with mixed-methods research or design science will be a plus, but not mandatory.

Inclusive ICT

Course Code	XM_0177
Credits	6
Period	P6
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K. Aslam

Examiner	dr. K. Aslam
Teaching Staff	dr. K. Aslam, dr. J. Gordijn
Teaching method(s)	Study Group, Lecture

Course Objective

At the end of the course, students will be able to:

1. Develop an understanding of the socio-economic characteristics of marginalized and minority populations in the global North and the disparities in ICT access and usage. (Knowledge and Understanding)
2. Elicit the specific requirements and limitations of marginalized populations and then propose tailored ICT solutions for them. (Applying Knowledge and Understanding)
3. Analyze the potential impact and ethical implications of the designed ICT solutions on the target populations. (Making Judgements)
4. Effectively communicate the outcomes of ICT needs assessments and solution proposals both in written reports and oral presentations. (Communication)
5. Engage in continuous learning about the evolving needs of marginalized populations in relation to ICT applications and maintaining an adaptive approach to problem-solving in these cases. (Lifelong Learning Skills)

Course Content

This course will provide students with an understanding of the global North's socio-economic landscape and the marginalized and minority populations within it (e.g., differently-abled people, migrants and refugees, elderly population, ethnic minorities etc.) exploring disparities in ICT access. The course objectives closely align with several of the United Nations Sustainable Development Goals (SDGs) such as Reduced Inequalities (SDG 10), Sustainable Cities and Communities (SDG 11), and Peace, Justice, and Strong Institutions (SDG 16).

We will discuss methods for assessing ICT needs for these populations through case studies and community engagement activities. Students will learn about designing inclusive ICT solutions also by working together in groups where they will propose and present a tailored ICT application. The course will also incorporate discussions on the effectiveness of public-private partnerships and policy implementations in fostering ICT accessibility in the Global North.

Additional Information Teaching Methods

The lectures will be delivered in an interactive seminar-style format. Prior to each class, students are expected to engage with assigned readings and multimedia resources to build a foundational knowledge base. During the lecture, students will also reflect on their own viewpoint in relevance to the material discuss in class.

Method of Assessment

Two group assignments: 60%
Four pass/fail homeworks (individual, may also include delivering recorded presentation): 30%
Attendance: 10%

Each of the assignments should be at least graded with a 5.5 or higher and the total marks should be 6 or higher for a student to pass the course.

Literature

Literature will be made available on Canvas.

Additional Information Target Audience

MSc CS, MSc AI, MSc IS

Knowledge Graphs and Semantic Technologies

Course Code	XM_0147
Credits	6
Period	P4
Course Level	400
Language Of Tuition	English

Faculty	Faculty of Science
Course Coordinator	dr. I. Tiddi
Examiner	
Teaching Staff	prof. dr. K.S. Schlobach, dr. I. Tiddi
Teaching method(s)	Practical, Lecture

Course Objective

1. Learn the requirements of representing knowledge in the form of graphs (Knowledge and understanding)
2. Master the RDF, RDFS, OWL and SPARQL languages, and their reasoning abilities (Apply knowledge and understanding)
3. Acquire dexterity in technologies that use these languages (NoSQL databases, RDF libraries, query languages, ontology editors, machine learning methods to learn from graphs) (Knowledge and understanding)
4. Apply these technologies to build fundamental Knowledge Graph infrastructure (Apply knowledge and understanding, Make judgments)
5. Design experiments on top of the built infrastructure to make a scientific contribution to the research field (Apply knowledge and understanding, Communication)
6. Obtain familiarity with the research fields of Semantic Web, Knowledge Representation, Information Extraction, and write a research paper about the identified research problems (Apply knowledge and understanding, communication, lifelong learning skills)

Course Content

In this course, you will learn :

1. the theory and practice methods to create structured knowledge in the form of graphs
2. the application of these methods to real-world problems and data
3. the research methods behind them.

Additional Information Teaching Methods

1. Content lectures (videos of knowledge fragments)
2. Practical sessions in class to work on Notebooks and Project (attendance mandatory)
3. Guest lectures in class (attendance mandatory)

Method of Assessment

A weighted average of:

1. Exam (35%) at least 5.5/10 to pass the course
2. Practical assignments (15%)
3. Group Project (35%)
4. Research Paper Analysis (5%)

Resit possible for the exam only.

Literature

1. Knowledge Graphs (A. Hogan et al., https://dl.acm.org/doi/abs/10.1145/3447772)
2. Handbook of Knowledge Representation (F. van Harmelen, V. Lifschitz, B. Porter)
3. A Semantic Web Primer (F. van Harmelen, G. Antoniou)
4. Linked Data: Evolving the Web into a Global Data Space (T. Heath, C. Bizer; http://linkeddatabook.com/editions/1.0/)

Additional Information Target Audience

Master Artificial Intelligence

Explanation Canvas

All announcements and materials are made available via Canvas

Recommended background knowledge

Python Programming basics (we mostly use RDFlib).

A plus:

- Knowledge of logics (SAT, constraint programming, description logics, qualitative reasoning)
- Knowledge of Machine Learning

Knowledge Organization

Course Code	X_405065
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. R.M. Siebes
Examiner	dr. R.M. Siebes
Teaching Staff	T. Kuhn, dr. R.M. Siebes, dr. V. de Boer
Teaching method(s)	Seminar

Course Objective

At the end of this course, students will:

- have gained high-level knowledge of the potentials and problems with formal knowledge organization (knowledge and understanding);
- have gained high-level knowledge of the possible solutions and applications for formal knowledge organization (knowledge and understanding);
- be able to understand concrete knowledge organization methods and practically apply them (knowledge and understanding, apply knowledge and understanding);
- be able to communicate own thoughts and reflections on the topic of knowledge organization (communication)

Course Content

This course covers the general principles and methods that form the foundation of information organization and knowledge-intensive processes, as well as the contexts in which they can be applied and the interaction with users. The lecture topics include knowledge modeling, ontologies, logic, controlled natural language, Semantic Web and Linked Data, as well as knowledge maintenance and evaluation, in addition to guest lectures on specific applications and domains.

Additional Information Teaching Methods

Lectures are in a hybrid setup and can be followed both in the lecture room and online, and will be video-recorded. These recordings are made available through Canvas afterwards.

Method of Assessment

Final written exam (75%), final assignment (25%). All parts are individual (no group work). To pass the course students need a passing grade (5.5 or more) both for the exam and for the final assignment. There will be a resit for the written exam and for the final assignment. For the final assignment, the maximum resit grade is 6.0.

Literature

Papers to read will be announced through Canvas.

Additional Information Target Audience

MSc Artificial Intelligence
MSc Computer Science
MSc Information Sciences

Additional Information

The course has a maximum enrollment limit of 120. The students of the Master Information Sciences, for whom it is compulsory, have priority over the students that take this course as an elective or a constrained choice course.

The students that take this course as an elective or a constrained choice course run the risk of not being able to participate, even if they have registered on time, if the maximum enrollment limit has been reached.

The students of the Master Information Sciences, for whom it is compulsory, run the risk of being initially refused if the enrollment limit has been reached at the time of registration. However this will be adjusted manually after registration has been closed so that all the students of the Master Information Sciences will be able to participate.

Knowledge Representation

Course Code	XM_0059
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. P.S.H. Koopmann
Examiner	dr. P.S.H. Koopmann
Teaching Staff	dr. P.S.H. Koopmann
Teaching method(s)	Practical, Lecture

Course Objective

Knowledge and understanding: at the end of the course the students should be acquainted with the broad principles of knowledge representation, such as the separation of representation and reasoning, the declarative nature of representations, the universal (domain independent) nature of inference mechanisms.

Apply knowledge and understanding: students will have practical experience with different representation formalisms, and will be able to implement a reasoning tool for at least one of these formalisms. This will allow them to better understand the role of knowledge representation in the broader context of AI.

Making judgement: students will be able to set up empirical experiments in order to evaluate the pros and cons of Knowledge Representation formalisms in specific application areas.

Communication skills: students will be able to write a scientific report about an original research question in a small group of students.

Learning skills: students will be trained in acquiring knowledge about a set of complex formal systems, learn how to come up with a research question and scientific hypotheses, and perform the necessary (empirical) research to prove or disprove those hypotheses.

Course Content

We discuss important formalisms for Knowledge Representation and symbolic AI, such as classical logics, description logics and one additional relevant formalism dealing with non-montonic and/or quantitative reasoning.

Additional Information Teaching Methods

Lecture: 2 sessions of 2 hours each week, on demand a Q&A session every 2-3 weeks, Working Groups to work on exercises every 2-3 weeks, a combination of exercises and practical assignments. The practical assignment will involve implementation, experimental evaluation and writing a report.

Method of Assessment

- assignments in groups (total 50%),
- peer reviewing assignment reports (P/F)
- written exam (50%).

A minimum grade of 5.5 for each component is required. The peer reviewing assignment report is marked pass/fail only, but must be completed to finish the course (even for resit students).

There will be a resit of the written exam, there is no resit opportunity for the practical assignments.

Literature

The main source of information for the lecture will be the lecture and the slides - no additional literature research is mandatory.

Additional Information Target Audience

Master Artificial Intelligence
Master Bioinformatics and Systems Biology

Recommended background knowledge

1. Basic knowledge of logic (propositional logic and first-order logic)
2. A good working knowledge of programming in Python is required

Language as Data

Course Code	L_PAMATLW001
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Humanities
Course Coordinator	dr. L.E. Donatelli
Examiner	dr. L.E. Donatelli
Teaching Staff	dr. L.E. Donatelli
Teaching method(s)	Seminar, Lecture

Course Objective

In this course, students will learn to explain, design, and execute a complete linguistic data analysis pipeline, transforming “messy” language into machine-readable input. The data analysis pipeline will integrate theoretical concepts from key areas of linguistics (morphosyntax, semantics, and pragmatics) with computational methods for NLP such as specific Python libraries, NLTK, spaCy, and Stanza that allow for targeted text processing and linguistic analysis.

By the end of the course, students will be able to:

1. identify and formulate theoretically grounded research questions;
2. collect, clean, and annotate linguistic data for a specific task;
3. apply computational techniques to model linguistic information in corpora; and
4. critically interpret results in light of both formal linguistic theory, standard NLP evaluation metrics, and social and contextual dimensions of language use.

Students will demonstrate their competence through hands-on assignments and a final project. Throughout, students will reason rigorously about linguistic data and representations, as well as standard methodological choices from both computational and theoretical perspectives.

Course Content

Language as Data introduces students to the linguistic and computational foundations necessary to analyze natural language as structured, meaningful, and context-sensitive data. The course focuses on how morphosyntactic structure, semantic content, and pragmatic context interact in real-world language use, and how these layers of information can be modeled and analyzed across corpora. Students learn to represent and manipulate linguistic data using techniques from natural language processing (NLP), while grounding their analyses in core linguistic theories of syntax, meaning, and use.

The course explores how linguistic patterns emerge across diverse domains, languages, and genres, and how such variation can be captured or abstracted in computational representations. By linking theoretical insight with hands-on computational practice, the course prepares students to engage critically with both academic research and real-world language technologies. Students will learn to identify how linguistic assumptions shape data analysis, to detect and mitigate biases in language data, and to design research that reflects both theoretical and computational best practices. Finally, students will learn to use and critically evaluate language models for research and general academic practice.

Additional Information Teaching Methods

The course consists of two sessions per week. In the lectures, the relevant linguistic theory and NLP methods are introduced and in the practical sessions, students learn to write code to apply the theoretical concepts and represent language as data. Students are expected to show an interactive attitude.

Method of Assessment

The course is evaluated by practical assignments (50%) and an exam (50%). Both components need to be graded at least with a 5.5. Depending on the number of registrations, practical assignments might be organized as group projects and the exam might be organized as an oral exam.

Entry Requirements

Students must have acquired programming skills in Python on the level that is taught in the course "Programming in Python for Text Analysis" and should have background knowledge in natural language processing on the level that is taught in the course "Introduction Human Language Technology".

Literature

will be communicated on Canvas

Additional Information Target Audience

The course is optimized for MA students in Linguistics (Language & AI specialization, and year 1 of HLT). Students from other masters with sufficient Python programming skills and NLP background knowledge are also welcome.

Explanation Canvas

Details about the course will be published on Canvas.

Learning Machines

Course Code	XM_0061
Credits	6
Period	P3, P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. K. da Silva Miras de Araujo
Examiner	dr. K. da Silva Miras de Araujo
Teaching Staff	dr. K.S. Luck, dr. K. da Silva Miras de Araujo, dr. A. Yaman
Teaching method(s)	Seminar, Lecture

Course Objective

1. Understanding and being able to critically discuss how learning machines relates to machine learning: This includes the main aspects involved in both, as well as their implications. (Knowledge and Understanding)
2. Designing systems of robots able to adapt to solve tasks: This includes endowing a robot with learning mechanisms that make use of sensing capabilities. (Applying Knowledge and Understanding)
3. Designing and conducting experiments with simulated and real robots to solve tasks. (Applying Knowledge and Understanding)
4. Being able to analyze the results from experiments (3), deriving insights into diverse aspects related to the methodology utilized. (Judgement)

Course Content

- This course concerns **designing robot systems** that can sense their environment, act upon it, and improve their behavior to solve tasks.
- **Most** of the course encompasses **practical sessions** with only a few lectures.
- Multiple tasks should be delivered, and students have the freedom to choose different learning and optimization methods to work with.
- All tasks will be done in **simulation AND in hardware** using Robobo robots. The Robobos are wheeled robots with fixed bodies that possess infrared sensors and a camera.
- Students are expected to **design/optimize the controllers (brains)** of the robots to solve the due tasks while utilizing the sensors as inputs.
- For the simulations, a robot programming/simulation framework will be provided to students. The **simulator** utilized is VREP (Coppelia).

Additional Information Teaching Methods

- The course is short and requires **intensive work** - it happens over a period of **four weeks** - each week contains one **lecture**, two **practical** sessions, and one **presentation** & evaluation session.

- **Attendance at all sessions is mandatory.** Missing lectures and practical sessions must be justified to the teacher and might be very detrimental to the assignment of the week, while missing a presentation incurs failing the assignment of that week.
- Students will work in **teams**.
- This is a course for students willing to go the extra mile, requiring a high level of **independence**.

Method of Assessment

1. Task I (presentation and demo): 20/100
2. Task II (presentation and demo): 20/100
3. Task III (presentation and demo): 20/100
4. Final report: 40/100

Tasks scores range from 0 to 10, and **all tasks must be above 5.5 to pass the course**

In each presentation, the group is required to disclaim each member's contribution explicitly. Additionally, each member must participate in presenting and, most importantly, in answering questions from the teacher or TA during/after the presentation. If the student does not **demonstrate enough individual knowledge**, their score for that assignment will be individually penalized.

The course does **not offer resits**, but students can **reuse assignment's grades** in case they fail and enrol again.

Additional Information Target Audience

Master Artificial Intelligence
Master Business Analytics

Recommended background knowledge

- Having passed any course involving **learning algorithms**, e.g., Reinforcement Learning, Evolutionary Computing, Computational Intelligence, Deep Learning, etc. - the course will not teach you new algorithms and it is fundamental that you have intermediate skills with at least one learning algorithm.
- Intermediate **programming** skills
- Python
- Independence for self-study

Not mandatory, but useful: Experience with robotics simulators - these can be very challenging to work with.

Machine Learning and Reasoning for Health

Course Code	XM_0102
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. A.C.M. ten Teije
Examiner	prof. dr. A.C.M. ten Teije
Teaching Staff	prof. dr. A.C.M. ten Teije, dr. A.J. Lisowska MBA
Teaching method(s)	Practical, Lecture

Course Objective

The main aim of this course is to make the student familiar with the combination of machine learning and reasoning techniques applied in healthcare. We will focus on machine learning with prior knowledge (“informed machine learning”), machine learning with symbolic output, and explainable learning systems. We apply each approach to some of the important tasks in healthcare, namely diagnosis, prediction, treatment and prevention using patient data (electronic health record data or trial data).

The course is focused on the following Bloom descriptors:

- Understand and apply a generic framework for machine learning (ML) and reasoning to solve real-world problems in healthcare.
- Demonstrate proficiency in designing and implementing machine learning models that incorporate prior domain-specific knowledge.

- Develop and evaluate machine learning models that produce symbolic outputs for healthcare applications
- Analyze and implement techniques to improve the interpretability and generalisability of machine learning systems
- Apply machine learning and reasoning techniques to healthcare applications, such as disease prediction, diagnosis, treatment planning and monitoring

Course Content

The lectures are organized into four modules corresponding to the stages of ML and reasoning solution development in healthcare:

- Module 1: Initiating a Healthcare ML Project
- Module 2: Design of Learning and Reasoning Systems
- Module 3: Evaluating the AI System
- Module 4: Adding to Existing Knowledge

Module 1: Initiating a Healthcare ML Project

Module 2: Design of Learning and Reasoning Systems(Machine learning with prior knowledge) The input of informed machine learning consists of usual training data and additional prior knowledge. The prior knowledge is independently of the learning task and can be in the form of logic rules, simulation results, knowledge graphs, etc.

Topics:

- Design patterns.
- Prediction with machine learning techniques (i.e., CART, LR, RF) enriched with domain knowledge (ontologies). For example predictive modeling of colorectal cancer using a dedicated pre-processing pipeline on routine electronic medical records.
- Reinforcement learning with initial policies based on domain knowledge and/or constraints. Reinforcement learning can be used to select sequences of interventions. This should however be done safely, and one should not ignore existing knowledge on suitable intervention strategies. For example, limiting the actions space per situation based on rules and regulations and starting from the current treatments regimes, and refining them based on data.

Module 3: Evaluating the AI system
Machine learning algorithms are usually black boxes, however physicians would like an explanation for the decisions of machine learning algorithms.

Topics: SHAPE and LIME algorithms for providing an explanation. Those algorithms can be used for instance for explaining predictions of breast cancer survival or using background knowledge for deductively reconstructing an explanation for the results of machine learning algorithms.

Module 4: Adding to existing knowledge(Machine learning with symbolic output) Machine learning techniques can be applied to symbolic structures (eg. Electronic Health Record) that results in symbolic output (e.g. discovery of possible causes of rare diseases, or simple guidelines).

Project: exploiting a combination of reasoning and learning in health care along the lines of the above themes (machine learning with prior knowledge, machine learning with symbolic output, explainable learning. systems). For instance using several machine learning techniques and knowledge sources to predict the occurrence of a particular disease based on medical records.

Additional Information Teaching Methods

Lecture series, plus workgroup sessions for the assignment. In the first part of the period the emphasis is on the theory (lectures), the second part of the period is a practical assignment in the form of a project.

The course ends with an exam.

Furthermore we aim to have one or two guest lectures by medical experts (physicians).

Method of Assessment

Written exam (E) (50%) and practical assignment (A) (50%). For both parts the grade needs to be sufficient to obtain a final grade. No resist is possible for the practical assignment.

Literature

Collection of research papers.

Additional Information Target Audience

Master students Artificial Intelligence (track AI for Health)

Master students Medical Informatics (UvA) (track AI for Health)

Custom Course Registration

Registration for medical informatics (UvA) students: Please checkthis document and then register for secondary courses via https://vu.nl/en/student/information-about-registration-and-enrolment

Machine Learning for Graphs

Course Code	XM_0119
Credits	6
Period	P3
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E. van Krieken
Examiner	dr. E. van Krieken
Teaching Staff	dr. E. van Krieken, prof. dr. D. Coumou, dr. P. Bloem, drs. W.X. Wilcke MSc, dr. V.M. Galfi
Teaching method(s)	Study Group, Lecture

Course Objective

Upon completion of this course, students

- Will be acquainted with several modern approaches to apply machine learning to problems that explicitly or implicitly involve graphs (Knowledge and understanding)
- Have deep knowledge of one or more selected methods, down to the details of the primary research (Knowledge and understanding)
- Are able to implement, extend, and evaluate a selected method (Applying knowledge and understanding; Make judgements)
- Can critically reflect upon the inner workings of a selected method and on the choices made by its authors (Learning skills)
- Can engage in a scientific discussion with fellow students about the pro's and con's of an arbitrary method, and is able to write a scientific reports of their findings (Communication skills)

Course Content

Graphs are one of the most powerful ways of representing data. They can naturally describe a wide range of relational constructs, including social connections, historical contexts, ways of moving through a city, and interactions between molecules in a cell. Multi-relational data, consisting of entities, (multi-modal) attributes, and the labelled relations between them, can likewise be encoded as a knowledge graph in a most natural fashion. Since much of the world's knowledge is of a multi-relational nature, the idea of knowledge graphs as the preferred data model has gained widespread adoption by technological giants such as Google and Facebook.

This widespread adoption has led to a fast growing interest in machine learning models for (knowledge) graphs, with recent years witnessing the development of various powerful deep learning techniques which enable machine learning researchers and practitioners to learn on, to learn from, and to learn with graphs directly and with minimal loss of information. These advancements have shown the potential that machine learning for graphs can bring us, while also having identified the challenges that still lie ahead.

In this course, we will focus on the bleeding edge of research on the intersection between machine learning and graphs. The course begins by revisiting the fundamentals of (spectral) graph theory as it pertains to machine learning. Next, we will focus on specific machine learning techniques for graphs, moving chronologically from established and proven methods to the more recent and cutting edge approaches, as well as discussing scalability, various downstream tasks, and learning on multi-modal heterogeneous knowledge. Finally, during the second half of the course, we will move our focus to reading and discussing recent peer-reviewed research papers.

Topics include:

- Embedding methods (Node2Vec, RDF2Vec, Distmult, RESCAL, translational methods, etc)
- Message passing methods and Graph networks (GCNs, GATs, R-GCNs, etc)
- Rule-based methods (AIMIE, AnyBURL, etc)

- Learning tasks (link prediction, node classification, representation learning, query and question answering, etc)
- Opportunities and challenges (scalability, mini-batch learning, multi-modal learning, etc)

Additional Information Teaching Methods

There will be five meetings per week, one on each day, consisting of either a (guest) lecture, a working session, or a presentation followed by a discussion. Students are assumed to not have other concurrent courses.

Lectures will be given during the first half of this course, introducing relevant background knowledge and presenting recent machine learning techniques for (knowledge) graphs. Two working sessions will be held during this period as well, providing hands-on experience with graph data and relevant machine learning tools.

During the second half of the course the format will change to a research seminar for which you need to read some of the latest literature in the field, form arguments about them, and then discuss these arguments during a scientific debate.

Students will also work on a practical project for most of the course. Weekly supervision meetings are provided, either on site or online.

The course is taught in English. Attendance during the seminar part is mandatory in order to pass.

Method of Assessment

There will be two moments of assessment:

1. Argumentative skills and participation during a discussion.
2. A scientific report, documenting the practical project.

The argumentative skills and the report are both graded, with the former counting for one-third of the final grade and the latter for two thirds. Both parts must be passed with a minimum of 5.5 to pass the course.

A resit opportunity is provided for the scientific report. There is no resit opportunity for the discussion.

Literature

There is no textbook. Reading material will be provided digitally in the form of peer-reviewed research papers from various related conferences and journals, including NIPS, AAAI, ICML, ICLR, IJCLR, ISWC, ESWC, ACL, and EMNLP.

Additional Information Target Audience

Master Artificial Intelligence (year 1)

Master Artificial Intelligence (year 2)

Custom Course Registration

Because of this intensive, participatory, and interactive seminar format, this course is limited to 60 students.

Only a registration for the main course itself is required to take part in all parts of the course.

Explanation Canvas

All course material will be made available through Canvas.

Recommended background knowledge

We require a strong background in machine learning and all topics that it implies, such as linear algebra, calculus, and probability theory. Familiarity with graph theory is recommended but not required.

An extensive experience with Python programming is a hard requirement. Practical experience with implementing machine learning models, preferably using pyTorch, is also strongly recommended. Ideally, students have taken the Machine Learning (X_400154), Data Mining Techniques (XM_40012), and/or Deep Learning course (XM_0083).

Machine Learning for NLP

Course Code	L_AAMPLIN024
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Humanities
Course Coordinator	prof. dr. A.S. Fokkens
Examiner	prof. dr. A.S. Fokkens
Teaching Staff	dr. P.J.M. Sommerauer, prof. dr. A.S. Fokkens
Teaching method(s)	Seminar

Course Objective

This course provides students with the means to conduct NLP research using machine learning. Students will learn: a) what the main machine learning technologies used in Natural Language Processing are b) how they work and how they can be used c) the methodologies for using these technologies in NLP research and d) how to represent linguistic data and what the impact is of choices in data representation. By the end of this course, students will be able to (1) name and describe the (working of) main machine learning technologies in NLP, (2) apply these technologies to specific NLP tasks (3) design a research environment where machine learning is used to solve an NLP problem, and (4) interpret and analyze evaluation results from machine learning experiments.

Course Content

Machine learning is a dynamic and active research field. The main goal of machine learning is to develop systems which can automatically solve different problems without being specifically programmed, i.e. by learning from the data. In this course, we will focus on the use of machine learning as a methodology for solving NLP tasks (e.g. pos-tagging, syntactic parsing, information extraction (named entity recognition)). We cover both `traditional' machine learning methods as the latest deep learning approaches. Representation of language as data plays a prominent role in this course.

This course distinguishes itself from other ML courses taught at this university, in its focus on analysing tasks and data representation and, of course, natural language. We will cover the experimental setup, running existing packages on new tasks and evaluation of overall results as well as error analysis. The course covers practical skills that can be useful in industry as well as in academia.

The course can be followed by any student with sufficient linguistic and programming knowledge. Note however that this course is explicitly not meant as an introduction to NLP or linguistics: knowledge of basic linguistic concepts is an explicit prerequisite.

Additional Information Teaching Methods

2 lectures of 2 hours, plus office hours and a feedback session. One focuses on machine learning algorithms, the other on linguistic properties and practical aspects. Active participation is needed to be successful for this course. For this reason, please note that attendance of the lectures is obligatory. The feedback session is highly recommended, but not obligatory. Office hours can be used in case of need.

Method of Assessment

Students hand in a portfolio of exercises carried out during the course and take a final exam. Both components need to receive a passing grade in order to pass the course (at least 5.5).

Literature

TBA

Additional Information Target Audience

This course is specifically designed for students in the Text Mining 1-year master. It can also be followed by Computer Science students (among others Business Analytics and AI) if they have sufficient knowledge of linguistics or are willing to invest (independent) research time to obtain this (materials are provided).

Recommended background knowledge

Sufficient knowledge of linguistics, programming at the end level of Programming in Python for Text Analysis.

Programming (Python, corresponding to the end level of `Programming Python for Text Analysis'), Linguistics,

Natural Language Processing/Human Language Technology (NLP/HLT). Students with some prior knowledge of machine learning may manage without prior knowledge of linguistics or NLP/HLT.

Machine Learning for the Quantified Self

Course Code	XM_40012
Credits	6
Period	P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M. Hoogendoorn
Examiner	prof. dr. M. Hoogendoorn
Teaching Staff	B. Borsboom, prof. dr. M. Hoogendoorn, V.G.M. Francois Lavet
Teaching method(s)	Practical, Lecture

Course Objective

The main aims of this course are to make the student familiar with specific machine learning techniques for quantified self and sensory data. Both the theory as well as the practical side of matters will be addressed. More specifically, the following aims are distinguished:

1. Understanding the domain of the quantified self and sensory data
2. Knowing and being able to apply techniques for outlier detection on sensory data.
3. Knowing and being able to apply techniques for feature engineering on sensory data.
4. Knowing and being able to apply clustering techniques related to sensory data.
5. Understanding the theory of supervised learning and its implications.
6. Learning how to apply non-temporal machine learning techniques to sensory data.
7. Knowing and being able to apply temporal machine learning techniques to sensory data, including deep learning techniques.
8. Knowing and being able to apply reinforcement learning in the application of the quantified self.

The course if focused on the Dublin descriptors Knowledge and understanding (learning new relevant algorithm in this domain); Applying Knowledge and Understanding (since the aim is also to apply the algorithms in practice); Making judgements (which technique is appropriate, how to best apply the techniques for a specific case); Communication skills (how to report on your approach, choices and your results), and Learning skills (able to find new relevant techniques, assess their suitability, etc.).

Course Content

The quantified-self refers to large-scale data collection of a user's behavior and context via a range of sensory devices, including smart phones, smart watches, ambient sensors, etc. These measurements contain a wealth of information that can be extracted by means of machine learning techniques, for instance for the purpose of predictive modeling. In addition, machine learning techniques can be a driver for adaptive systems to support users in a personalized way based on the aforementioned measurements. The type of data does however require specialized machine learning techniques to fully exploit the information contained in the data. Examples of challenges include the temporal nature of the data, the variety in the type of data, the different granularity of various sensors, noise, etcetera.

In this course specific techniques to handle quantified self (or broader sensory data) will be treated. More in specific, it will address:

- Feature engineering (how do we come from raw data to usable features):
  * Removing noise from data
  * Handling missing data
  * Identifying (temporal) features
- Learning of user patterns:
  * Temporal machine learning approaches such as recurrent neural networks, time series analysis
  * Clustering approaches with dedicated distance metrics (including dynamic time warping)
- Adaptive feedback and support
  * Reinforcement learning
- Integration of the various components.

In addition, a number of real-life applications will be discussed. Next to lectures, there will be an extensive practical part, where students will learn to work with various algorithms work on a project they propose themselves.

Additional Information Teaching Methods

The course will be taught in four weeks. Both regular lectures (l, first two weeks) and practical sessions (p) will be organized. During the first two weeks the emphasis will be on the lectures and getting started with the practical assignment that ties closely to the content of the lectures. The third week will be working on the practical assignment fulltime. In the fourth week a mandatory final lecture will take place in which a selection of groups will present their project. The course ends with an exam.

Method of Assessment

Written exam (50%) and practical assignment (50%). For both parts, the grade needs to be sufficient to obtain a final grade. The practical assignment is done in groups of three. For the practical assignment, two intermediate versions of the report should be submitted for feedback that are given a pass/fail. Only in case of a pass for the two intermediate versions will a final grade be given for the practical assignment.

The written exam will be multiple choice and have a resit during the Summer period. On top, in case your assignment grade average is insufficient you are allowed to redo the assignments by improving it based on the feedback you received. The maximum grade for this resubmitted assignment is set to a 6.

Literature

Hoogendoorn, M. and Funk, B. Machine Learning for the Quantified Self - On the Art of Learning from Sensory Data, ISBN 978-3-319-66308-1, Springer Verlag 2018. The book can be downloaded free of charge through the UBVU, just follow the link to the book: http://dx.doi.org/10.1007/978-3-319-66308-1

Papers on state-of-the-art deep learning techniques.

Additional Information Target Audience

Master Artificial Intelligence, Master Business Analytics, Master Computer Science

Additional Information

Lecturer: Prof.dr. M. Hoogendoorn

Custom Course Registration

Registration for medical informatics (UvA) students: Please check the following document https://www.dropbox.com/s/a2kd7xacuxt5y8e/AI%20for%20Health%202022-2023.
and then register for secondary courses via https://vu.nl/en/student/information-about-registration-and-enrolment/in

Recommended background knowledge

Programming experience. Knowledge of basic machine learning algorithms.

Master Project AI

Course Code	X_400285
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. N. Ullah
Examiner	S.M.A. Zaghow
Teaching Staff	prof. dr. A.C.M. ten Teije
Teaching method(s)	Lecture

Course Objective

The Master program in Artificial Intelligence is a scientific program that aims to provide the student with the knowledge, experience and insights in AI needed to autonomously carry out his/her professional duties. It is designed to prepare the student for further education as scientific researcher (Ph. D. studies) as well as to offer a solid basis for a career in business at an academic level. Moreover, the program aims at educating students to acquire a practical understanding of the position of the field of Artificial Intelligence within a broad scientific,

philosophic and societal context. The Master Project AI marks the end of the Master program in Artificial Intelligence and aims to cover all its intended final learning outcomes.

The learning outcomes of the course (based on the Dublin descriptors) are as follows:

1. Knowledge and Understanding (reading up on relevant AI techniques for the project at hand);
2. Applying Knowledge and Understanding (applying the AI techniques within the project);
3. Making judgements (finding an appropriate technique, applying the technique for a specific case);
4. Communication skills (reporting on your approach, choices and your results and how to present your work)
5. Learning skills (e.g. finding new relevant techniques, assessing their suitability, etc.).

Course Content

The Master Project AI marks the end of the Master programme in Artificial Intelligence. Information about possible projects (incl. internships) can be found on the Master Projects Canvas site.

Internships proposed by students need prior approval by a staff member, who will also be involved in the project supervision.

The graduation project lasts between 5 and 6 months, and (typically) involves implementation of an artificially intelligent system and conducting experiments to evaluate performance. Deliverables include (not limited to) a project plan written in the first month, a written thesis and a final presentation.

For more details and deliverables, see Canvas and Assignments.

Additional Information Teaching Methods

Your master project is typically supervised by a staff member with a PhD title and (general) expertise in the topic of your research. Postdocs and (assistant/associate) Professors are hence qualified to be your main supervisor. Persons outside of the VU AI staff may also be asked as supervisor. A PhD candidate may still be your daily supervisor, in which case the main supervision should be in the hands of someone who does have a PhD title. The grading is performed by two examiners who hold a PhD degree, of which at least one should be a VU staff member (typically the supervisor). The second examiner needs to comply with the VU's formal requirements for second examiners as well, even in the case that (s)he is not a staff member at the VU or a collaborating University.

Method of Assessment

The final grade will be based on the quality of the research (33.33%), attitude and development (16.7%), the written thesis (33.33%), and the final presentation (16.7%).

For detailed information on grading see the grading rubric on Master Project AI Canvas page.

Entry Requirements

The student needs to have successfully completed 60 ECTS (in total) of the MSc Artificial Intelligence to start with the master thesis, which marks the final part of the master AI.

Additional Information Target Audience

Master Artificial Intelligence

Additional Information

For all rules, assessment criteria, contact persons, and many practical tips for your master project, see the Master project page on Canvas.

Master Thesis: Research Project Cognitive Science

Course Code	P_MTHRCSC
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	prof. dr. H.A. Slagter
Examiner	prof. dr. H.A. Slagter

Teaching Staff	
Teaching method(s)	

Course Objective

To learn how to perform research in cognitive psychology / cognitive (neuro)science and report its results in a Master thesis. Projects involve basic research, applied research, research concerning modeling, or a combination of these.

Course Content

Students participate in a research project concerning cognitive psychology / cognitive (neuro)science. The project will be supervised by a member of the section Cognitive Psychology (part of the Department of Experimental and Applied Psychology at the Faculty of Behavioral and Movement Sciences). Before starting, a written research plan (half a page outlining the research question and the methods indicating how you intend to address that question) should be submitted to the coordinator of this course.

Participation in a research project can only start after approval of the research plan. The research performed by the student forms the basis for the thesis. The thesis should be written in journal article style at a level appropriate for submission to an academic journal. Students will be supervised by a person from the academic staff of the section Cognitive Psychology.

There will be at least one meeting a week between the student and the supervisor.

Additional Information Teaching Methods

Individual meetings with supervisor.

Method of Assessment

The final grade for the Master Thesis will be based on the quality of both the student's performance during the research and the quality of the written thesis. Grading will be done by the supervisor and a second assessor. It is required that students present their research in the form of a talk during one of the KIM meetings organized by the Department of Artificial Intelligence.

Additional Information Target Audience

AI students in the cognitive science track that do their thesis with a researcher outside of the Cognitive Psychology department should register for the Master Project AI course instead.

Medical AI

Course Code	XMU_0038
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	S.M.A. Zaghow
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

This is an UvA course. Please follow this link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026/zoek-vak/vak/124912

Additional Information Teaching Methods

-

Method of Assessment

-

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Medical Informatics Basics

Course Code	XMU_0037
Credits	6
Period	P5+6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	S.M.A. Zaghow
Examiner	
Teaching Staff	
Teaching method(s)	

Course Objective

This is an UvA course. Please follow the link for the course description.

Course Content

https://studiegids.uva.nl/xmlpages/page/2025-2026-en/search-programme/programme/8822/269403

Additional Information Teaching Methods

-

Method of Assessment

-

Custom Course Registration

This course is offered by the UvA. For more information contactFNWI Education Service Centre.

Enrolment via https://glass.uva.nl is required.

Memory and Memory Disorders

Course Code	P_MMEMORY
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	dr. R.J. Godijn
Examiner	dr. R.J. Godijn
Teaching Staff	
Teaching method(s)	Lecture

Course Objective

The course aims to give students an overview of memory at the cognitive and neurophysiological level, and to give students the background to interpret memory disorders in patients with brain damage.

Course Content

The course focuses on various approaches in the study of human memory and memory disorders. We will discuss different memory systems, retrieval, forgetting, eyewitness testimony, memory in childhood and aging and the brain substrate of memory. We will also discuss memory loss after local brain damage.

Additional Information Teaching Methods

Short background lectures followed by paper discussions

Method of Assessment

Exam (50 %), presentation (20%) and research proposal (30 %).

Literature

papers on canvas

Additional Information

This course is taught every two years. Will be taught in 2024-25, but not in 2025-26

Multi-Agent Systems

Course Code	XM_0052
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E.J.E. Pauwels
Examiner	dr. E.J.E. Pauwels
Teaching Staff	prof. dr. K.V. Hindriks, dr. E.J.E. Pauwels
Teaching method(s)	Seminar, Lecture

Course Objective

After successfully completing this course, the student

- Has a solid understanding of concepts from elementary and intermediate game theory, such as Nash equilibria for simultaneous games, backward induction and subgame perfect equilibria for sequential games and the Shapley value in cooperative games.
- Understands various principled approaches to balance exploration and exploitation;
- Has a solid understanding of the tabular solution methods for (single agent) reinforcement learning;
- Is able to explore and digest current research on deep reinforcement learning and multi-agent reinforcement learning.

Dublin descriptors:
1 Knowledge and Understanding,
2. Applying Knowledge and Understanding:

Course Content

In Multi-agent systems (MAS) one studies collections of interacting, strategic and intelligent agents.

These agents typically can sense both other agents and their environment, reason about what they perceive, and plan and carry out actions to achieve specific goals. In this course we introduce a number of fundamental scientific and engineering concepts that underpin the theoretical study of such multi-agent systems. In particular, we will cover the following topics:

- Agents: Typology and examples, strategic versus learning agents
- Introduction to non-cooperative game theory

- Introduction to coalitional game theory for teams of selfish agents
- Principles of Mechanism Design
- Exploration versus Exploitation
- Markov Decision Processes
- Reinforcement learning for a single agent
- Introduction to multi-agent reinforcement learning

Additional Information Teaching Methods

Two lectures (1h45) and one recitation class (1h45) per week.

Method of Assessment

There will be weekly homework assignments that will be graded (5 of which are done in groups and 1 is done individually). In addition, there will be a final exam that will test the student's ability to apply the course material to new and concrete problems.

The final grade will be a weighted average of the grades for the homework assignments (50%) and the final exam (50%).

There is a resit offered for the final exam and the individual homework assignment. There is no resit for the group homework assignments.

Literature

Recommended reading:

Yoav Shoham, Kevin Leyton-Brown: Multiagent Systems
Publisher: Cambridge University Press (15 Dec. 2008)
ISBN-10: 0521899435
ISBN-13: 978-0521899437

R.S. Sutton, A.G. Barto, F. Bach: Reinforcement Learning: An Introduction
Publisher: MIT Press; second edition edition (23 Nov. 2018)
Language: English
ISBN-10: 0262039249
ISBN-13: 978-0262039246

Additional Information Target Audience

Master Artificial Intelligence

Master Business Analytics

Recommended background knowledge

Basic calculus, probability theory and linear algebra. Fluency in a programming language.

Natural Language Processing

Course Code	XM_0121
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. L.E. Donatelli
Examiner	dr. L.E. Donatelli
Teaching Staff	dr. L.E. Donatelli
Teaching method(s)	Seminar, Lecture

Course Objective

Natural language processing (NLP) is a highly dynamic research field that aims at analyzing natural language using computational models and methods. This course provides an introduction to the principles of the field and has two

main objectives:

1) Students learn the fundamentals for developing a system for a common NLP task.

- They can analyze and interpret the linguistic information provided by common pre-processing pipelines and are familiar with the core NLP terminology. (Knowledge & Understanding)
- They can describe a common NLP classification setup and evaluate the results. (Making Judgements)
- They can explain the functionality of neural networks and their use in NLP. (Communication)
- They can explain the role of pre-trained language models in NLP and apply them to a task. (Applying knowledge and understanding)

2) Students learn to systematically analyze and interpret NLP models for a specific task.

- They can describe the linguistic challenges associated with the task. (Making judgments)
- They can describe how the task is approached by a state-of-the-art model. (Knowledge and understanding)
- They are able to understand, evaluate, and interpret the output of a model. (Applying knowledge and understanding)
- They can compare the output of different models and analyze their strengths and weaknesses. (Lifelong learning skills)

Course Content

Natural Language Processing operates on the interface between linguistics and computer science. In order to get computers to deal well with natural language, it is important to understand both how language works and how computational methods work. Computational linguists work on this interface and have developed methods and technologies for language analysis.

This course covers technologies and computational models for core domains of natural language processing. Students are trained to find, process, and understand the latest developments in this rapidly advancing field. The course includes practical components that require students to work with Python code to explore NLP models. They learn to apply them to new data and reflect on the effect of modeling decisions on the performance.

Additional Information Teaching Methods

The course consists of two sessions per week. We generally alternate between more theoretical lectures and interactive practical classes in smaller groups.

Method of Assessment

The course is evaluated by practical assignments (50%) and an exam (50%). Both components need to be graded at least with a 5.5. Retakes of practical assignments to improve the grade are not allowed. Practical assignments might be organized as a mix of group projects and individual contributions.

Literature

This information will be specified on Canvas.

Additional Information Target Audience

Master Artificial Intelligence, also suitable for Master Business Analytics and other Master students in Computer Science

Recommended background knowledge

Students must have acquired programming skills in Python. For students with no background knowledge of machine learning, additional reading might be required as the fundamental concepts will only be briefly introduced in the course.

Neural Models of Cognitive Processes for AI

Course Code	P_NEUMOD_AI
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Behavioural and Movement Sc.

Course Coordinator	dr. T.H.J. Knapen
Examiner	dr. T.H.J. Knapen
Teaching Staff	dr. T.H.J. Knapen, mr. S. Zhang MSc
Teaching method(s)	Computer lab, Lecture

Course Objective

Computational modelling is an important tool for cognitive neuroscience. Conversely, neuroscience has proven to be a very rich source of inspiration for AI. This course will teach you the background knowledge on the core principles of computational modelling, and dive into the links between AI and (cognitive) neuroscience.

The course is intended to offer insight(s) into what different types of models exist in in cognitive neuroscience, how they can be (and are) used to enrich the field, and it explores what questions arise when evaluating modelling work in this field. We will also investigate the parallels between current issues in computational cognitive neuroscience and open questions in AI.

Course Content

Computational models are an important tool in cognitive neuroscience. When used appropriately, a model allows for the integration of findings from a wide range of experiments. Rather than merely verbal theories, computational models are rich in detail and allow for a mechanistic view on how the brain produces its behavior.

An old adage from statistics is that "all models are wrong, but some models are useful". They are wrong because a model by definition is a simplification of reality, but they are useful when they generate testable predictions. However, it can be difficult to assess whether a model is too much of a simplification, and whether its predictions actually are useful. What makes a model good or bad? To what extent do models need to fit the data? And if multiple models fit the data, how do we choose which is the `better one'? In addition, modeling papers can at times seem rather enigmatic, and for the untrained reader it is all too easy to get lost in the mathematical equations that make up computational models.

This course takes a learn-by-example approach to give an overview of different modeling approaches that are common in neuroscience. We will start at a high level of abstraction, with models that are used to mathematically describe experimental data, with relatively little regard for their implementation in the brain. By means of practical sessions, you will get hands-on experience with some of these models and see how they are implemented. By means of debates, you will learn how to assess different models in terms of their strengths and weaknesses. Lastly, you will listen to neuroscience/AI podcast episodes, and present one highlighted researcher's work in class.

Additional Information Teaching Methods

Lectures and discussion, computer tutorial and practicals.

Method of Assessment

Grades are based on a weighted average of performance on a final exam and quizzes (75%), and class participation in the debate sessions and reading discussions (Perusall 15%, Podcast presentations 10%)

Entry Requirements

There is no explicit required knowledge. However, as the practicals have you work with Python code, it might be useful to familiarize oneself with the language. https://www.codecademy.com/learn/python offers a wonderful free online tutorial

Literature

Literature (articles, tutorials) will be provided through Canvas & Perusall.

Prevention of Mental Health Problems

Course Code	AM_470840
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. N. van den Bogerd

Examiner	dr. N. van den Bogerd
Teaching Staff	dr. N. van den Bogerd
Teaching method(s)	Computer lab, Study Group, Lecture

Course Objective

- You can explain the most important theoretical and scientific concepts in the field of prevention of mental health problems.
- You can demonstrate a basic understanding of methods to identify who is at risk for developing mental health problems.
- You can discuss the relevance of different risk and protective factors for mental health problems.
- You can describe relevant prevention programs for different mental health problems and the evidence for its effectiveness.
- You can conduct a systematic and critical review of the scientific literature in the field of mental health prevention programs and report on it.
- You can integrate gained knowledge on mental health prevention to make recommendations for further research and practice.

Course Content

For Health Science students this course fits in the program of the specialization Prevention and Public Health. Within this specialization you are trained to become a health promotor who is able to work in a theory- & evidence-based way and is able to link research, practice and policy.

For Psychology students this fits in the program of the Clinical Psychology specialization. Within this specialization you are trained to become a psychologist specialising in either the research, policy or practice of mental health care. Most courses in this specialization can be freely chosen and are all specific subtopics in mental healthcare, usually aimed at specific disorders or types of treatment.

Theoretical backgrounds of the prevention of mental health problems will be discussed, as well as currently used methods in preventive mental health care. Guest lecturers who work in the field of preventive mental health care will discuss current programs aimed at preventing several psychological symptoms and disorders. Also, the most important results of research conducted in the field of preventive mental health care will be presented. There will also be a focus on the implementation and evaluation of mental illness prevention programs. In the practicals students will tackle a self-chosen problem within the field of preventive mental healthcare, writing a report on it and presenting their most important recommendations.

All lectures and seminars will be in English. All examination is in English as well.

Additional Information Teaching Methods

This course is rewarded with 6 ECTS and runs in P3. Prevention of Mental Health Problems is a full-time course, this means that 40 hours a week are necessary to pursuit the goals of this course. Regular attendance during the weeks is mandatory. Teaching activities include: lectures and work group meetings.

Method of Assessment

An individual written examination that accounts for 50% of the final grade of this course.

A written assignment and presentation of the written assignment that together account for 50% of the final grade of this course.

To pass this course you have to have at least a 5.5 for the individual exam, the presentation and the assignment.

Entry Requirements

At the start of this course, we expect you to have mastered knowledge, insight, attitude and skills at a level which is comparable to the final qualifications stated by the Bachelor of either Health Sciences, Psychology or Artificial Intelligence at the VU.

Literature

A list of scientific papers to study will be listed on canvas.

Additional Information Target Audience

Health Science, Psychology and AI students.

Additional Information

This course will be offered for the final time within the MSc Health Sciences in the 2025–2026 academic year.

Custom Course Registration

Registration for this course via VU-net. Post-placement for this course is not possible.

Recommended background knowledge

- Basic knowledge of psychopathology (symptoms of the most common psychiatric disorders).
- Basic knowledge on what prevention programs are and how they are Developed

Project Reinforcement Learning

Course Code	XM_0120
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	V.G.M. Francois Lavet
Examiner	V.G.M. Francois Lavet
Teaching Staff	V.G.M. Francois Lavet, prof. dr. M. Hoogendoorn
Teaching method(s)	Study Group, Lecture

Course Objective

Knowledge and understanding: students will be familiar with basic knowledge of (deep) reinforcement learning and they will be able to apply them to sequential decision-making tasks under uncertainty, including in the context of relatively high-dimensional inputs (e.g. time series).

Applying knowledge and understanding: students will be able to understand how to implement deep RL algorithms, how to compare them and how to apply them in real life problems.

Making judgements: students will have an understanding of the key learning techniques for sequential decision-making tasks used in RL, as well as an overview of a few key AI applications for such techniques along with a short discussion on societal impacts.

Communication skills: students will be able to write a scientific report about their project (in a group of 2-3 students).

Learning skills: students will be trained in acquiring a set of complex AI and programming skills in a restricted period of time. They will also be trained in applying those techniques on a given problem and report adequately the results.

Course Content

Reinforcement Learning is a set of techniques that can be used to solve sequential decision-making tasks. When combined with deep learning as a function approximator, these algorithms are referred to as deep RL and they have been able to solve a variety of tasks that were previously out of reach for a machine (e.g. the game of Go). Thus, deep RL opens up many new applications in domains such as healthcare, robotics, smart grids, finance, and many more.

This course will provide an introduction to how real-world problems can be formalized as sequential decision-making problems (MDP, POMDPs, meta RL) as well as how (deep) reinforcement learning techniques such as deep Q-networks and actor-critic methods can be useful. The course will provide the theoretical foundations as well as an introduction to the practical skills necessary to apply (deep) RL techniques. Students will also develop their coding skills by developing (deep) RL agents.

Additional Information Teaching Methods

The course will be centered on the practical task of designing intelligent agents that can solve a sequential decision-making problem close to real-life (e.g. optimizing the use of a damp by storing/releasing energy on the

electricity grid, given some past time series of the daily electricity price).

For the first week, there will be the equivalent of 4 hours of online pre-recorded lectures, one/two practical sessions (1h30 each), one or two Q&A sessions in a flipped classroom style (1h30 each). All students should know the basic elements of RL and deep learning by the end of the first week.

During the second week, there will be 4 lectures (each 1h30) covering specific topics of (deep) reinforcement learning as well as one/two practical session(s) and a test for the theory part.

The last two weeks will focus on implementing RL algorithms to solve the project and write a scientific report describing the results obtained. This part also encompasses a significant amount of self-study to familiarise oneself with the (deep) RL algorithms and their implementations.

Method of Assessment

The final grade takes the following elements into account:

Participation to the group work (seminar group) sessions via the attendance sheet (10% of the grade).
Test (10% of the grade)
|Intermediate presentation (10% of the grade)
Report+code and final presentation (70% of the grade)

A resit for the report+code and final presentation (70% of the grade) is possible but there is no resit possible for the other parts. An individual component in the grade is included in the final presentation and should be passed by each student in order to pass the course.

Literature

- Sutton, Richard S., and Andrew G. Barto. Introduction to reinforcement learning. Vol. 135. Cambridge: MIT press, 1998. Recommended, but not compulsory. There will be a reader.
- François-Lavet, Vincent, et al. "An Introduction to Deep Reinforcement Learning." Foundations and Trends® in Machine Learning 11.3-4 (2018): 219-354.

Recommended background knowledge

The minimal required background is basic machine learning concepts (supervised learning, the concept of loss functions, etc.). Some notions about Markov Decision Processes (MDPs) and practical knowledge of deep learning are also strongly recommended.

Secure and Trustworthy Machine Learning

Course Code	XM_0179
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. ir. M.E. van Dijk
Examiner	prof. dr. ir. M.E. van Dijk
Teaching Staff	prof. dr. ir. M.E. van Dijk, M. Chen
Teaching method(s)	Study Group, Lecture

Course Objective

We will discuss various topics in the intersection of security and machine learning. The main goal of this course is to have you think critically about how and to what extent security guarantees can be given for machine learning tasks. The main objective is to study trustworthy intelligence augmentation from different perspectives.

Upon completion of this course, you will:

- Have knowledge about security problems that arise in machine learning (K&U, AK&U, C)
- Understand (mathematical) technique(s) towards remediating security problems (K&U, AK&U, C)
- Have knowledge of example(s) about machine learning for security (K&U)
- Have acquired critical thinking around machine learning and security (MJ, LS)

The acronyms above refer to the Dublin Descriptors covered by each learning objective, specifically: Knowledge

and Understanding (K&U), Applied Knowledge and Understanding (AK&U), Making Judgment (MJ), Learning Skills (LS), and Communication (C).

Course Content

The course touches on various topics in machine learning and security. Examples of topics that can be included are: adversarial machine learning for image classifiers; Differential Privacy for distributed Stochastic Gradient Descent; secure multi-party computation for federated learning; the use of ML in security such as intrusion detection, analysis of silicon Physical Unclonable Function designs, GAN models generating synthetic data such that collected personal data can be discarded, and the moral character of cryptography, security, and artificial intelligence.

Additional Information Teaching Methods

Lectures (105 minutes 2x each week for 4 weeks); Group Q&A (135 minutes 1x each week for 3 weeks) during which (a) we cover lecture material for which we had no time during lectures, (b) upon student request re-explain topics and/or discuss upcoming assignments and (c) answer student questions about the lectured material and problems.

Method of Assessment

(1) At the start of the course you will write a short (but at least 1/3 page, single column) essay on your study strategy and your grade expectation:

You cannot pass the course if you don't submit by the deadline. And those who don't simply get NS "No Show" as the final grade.

(2) Questions and problems will be posted for each lectured topic. At the end of the course there will be a final in a controlled environment where a subset of the posted questions and problems for each topic will be asked. You will be able to prepare for this final by collecting and learning answers throughout the course. The final tests whether you have acquired a sufficient overall understanding. The final has a pass/fail grade.

If you do not pass the final, then you fail the course. If you pass the final, then your final grade is determined by your individually graded assignments as described below.

There is no re-sit opportunity for the final. A grade for the final from a previous year does not transfer to this year.

(3) There will be six (A = 6) individually graded assignments in the form of research surveys/essays together with your own self-evaluations of your submitted works. Each assignment is assessed based on the quality, correctness, and completeness of the solution(s) or written exposition. Each assignment is graded as a Fail (F), Pass (P), or Excellent (E).

(An F is also given for an assignment if you do not meet its deadline for submission.)

The following explains in text a rather complicated and intuitive formula for computing the final grade. A visual picture will be presented at the start of the course (the editor does not allow it to be inserted here).

We set

A = # Assignments

and we compute

N = 2*(#P) + 3*(#E),

where #P indicates the number of Passes you have received and #E the number of Excellent. (Notice that A = #P + #E + #F.)

If you have submitted your study strategy essay by the deadline and if you passed the final, then your final grade is equal to

14/6 + (17/6)*(N/A) rounded to the nearest integer in {1,2,3,...,10}.

The specifics of this grading policy are subject to change if the need arises.

There are no re-sit opportunities for any of the assignments/essays. Grades for assignments from previous years do not transfer to this year.

(4) At any moment during the course the instructor can request a student to explain one or more of his/her assignments in order for the instructor to verify whether the student understands the material presented in the assignments; if the student is not able to explain his/her understanding in an assignment, then the grade for the

assignment will be downgraded to a fail.

Entry Requirements

The course attempts to be self-contained and explains the necessary machine learning concepts as well as security and crypto techniques. Nevertheless, students are required to be independent and study extra where their mathematical background is limited. For example, you will want to know how to compute gradients, know about the structure of a proof, can read mathematical notation, etc.

Literature

Slide decks, papers, and reference material where needed.

Additional Information Target Audience

Master Computer Security; Master Computer Science; Master Artificial Intelligence

Seminar Cognitive Neurosciences

Course Code	P_MSEMCNS_AI
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Behavioural and Movement Sc.
Course Coordinator	prof. dr. H.A. Slagter
Examiner	prof. dr. H.A. Slagter
Teaching Staff	prof. dr. H.A. Slagter
Teaching method(s)	Lecture

Course Objective

After this course, the student

- can describe central contemporary debates in cognitive neuroscience
- can explain, critically evaluate, and present the main ideas in these debates, their strengths and weaknesses, methodological implications and empirical support
- can apply these ideas to research findings and clinical conditions and to design new experiments
- can formulate ideas about, and discuss how the field of cognitive neuroscience should progress

Course Content

This is an advanced course that presents recent developments in the field of cognitive neuroscience, with a focus on overarching ideas. Why do we have a brain to begin with? What is the relationship between cognition and action? Can psychiatric disorders be reduced to brain dysfunction? And can artificial systems ever truly think and learn like humans? The course provides an overview of important cognitive and neuroscientific debates with a focus on influential theories, their strengths and weaknesses, their methodological implications and their empirical support. We will also explore what are important future directions for the field of cognitive neuroscience.

Additional Information Teaching Methods

Lectures, literature study, group assignments, and discussions.

Method of Assessment

Group assignments (30%) and a final exam (70%). Both the exam and the assignments (average) will have to be graded with at least a 5.5 to pass the course.

Entry Requirements

The requirement to participate is the completion of the basic Cognitive Neuroscience and Neuropsychology course. Alternatively, students may study the required literature by self-study.

Literature

Scientific articles. The list of articles will be announced on canvas (annually updated).

Skills for AI

Course Code	XM_0077
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. S.M. van Otterloo
Examiner	dr. S.M. van Otterloo
Teaching Staff	dr. S.M. van Otterloo
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

Skills for AI is a brush up course for students with limited background in mathematical skills necessary for understanding Artificial Intelligence: linear algebra, logic and probability.

The goals are:

- Familiarise students with logical and mathematical language (Knowledge and understanding)
- Let students use logic and linear algebra to express AI applications (Knowledge and understanding)
- Let students solve mathematical exercises, both manually and with the help of the computer in Python (Applying knowledge and understanding)

Course Content

This brush up course contains the following topics: logic, linear algebra, probability theory, and gradients and derivatives.

Logic-part:
The logic part focuses on propositional logic and first order predicate logic:

- For propositional logic, the following is covered: truth tables, boolean operators, semantic equivalence, rewriting rules, functional completeness.
- For first order logic, students will learn predicates, quantifiers, logic models, reasoning rules and converting natural language to first order logic.

Linear algebra:
The main topics are: systems of linear equations, linear (in)dependence, linear transformations and matrices, matrix operations, determinants and eigen vectors. The theory will be used to model computer science and AI problems such as computer graphics, language recognition, path finding and neural networks.

For probability theory: events, probability, conditional probability and Bayes' Theorem will be covered.

Gradients and derivative: Definition and interpretation of the derivative as a gradient, computing derivatives for basic polynomial functions.

Additional Information Teaching Methods

Lectures, and practical sessions

Method of Assessment

The course will be examined in the following way:

- Quizzes after each lecture on canvas. These determines 20% of the grade
- Assignments will be made in small groups of 2-4 students (making report using a given dataset and open questions. mostly python based): These determines 30% of the grade
- A final written exam at the end of the course. This determines 50% of the grade A minimum grade of 5.5 for each component is required.

A resit option for the final written exam will be offered. If a student misses of fails one project assignment, one assignment can be resubmitted in the week after the final written exam.

Literature

Part 1 Logic: All course materials will be provided via Canvas.

Part 2 Linear algebra: All course materials will be provided via Canvas.
Optionally, students can use the book: (Linear Algebra and its Applications, by David C. Lay, Steven R. Lay en Judi J. McDonald, global edition (fifth edition), Pearson.)

Students may wish to consult: (How to Think Like a Computer Scientist, Learning with Python, 2nd Edition, by Jeffrey Elkner, Allen B. Downey, and Chris Meyers) see the URL:
http://openbookproject.net/thinkcs/python/english3e/index.html

Additional Information Target Audience

Only first year master Artificial Intelligence students without an Artificial Intelligence/Computer Science bachelor degree can attend this course.

Recommended background knowledge

Some programming experience with python is necessary and jupyter notebooks is recommended.

Social Robotics

Course Code	S_SR
Credits	6
Period	P4
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Social Sciences
Course Coordinator	dr. D.F. Preciado Vanegas
Examiner	dr. J.F. Hoorn
Teaching Staff	dr. J.F. Hoorn, dr. D.F. Preciado Vanegas
Teaching method(s)	Study Group, Lecture

Course Objective

After completion of this course, the student is able to:

- Understand and discuss current thematic and research issues in social robotics;
- Critically discuss the role of theory building (i.e. formal modeling) within social robotics, and reflect on ethical and normative issues in robot research and design;
- Apply a critical-reflective attitude about up-to-date and interdisciplinary scientific research in the field of social robotics;
- Develop a design proposal and related research to tackle a communication problem in social robotics.

Course Content

The future is now and we are designing it. Certain layers of society already work with social robots on a day-to-day basis. Until recently, robots seemed to be the realm of engineering but with the rise of Big Data analysis and embedded software in the Internet of Things, Communication Science has a new role to play in understanding how digital communications (i.e. patterns, trends, read-outs) should be translated back into human-digestible forms. Social robots seem to be the ultimate interface between the digital and analogue world but in what way are robots different from humans? Do the same rules of conduct and communication apply as to human-human interactions? Will Computer-Mediated Communication branch off into robot-mediated communication? Will Media Psychology have found a new field to explore human bonding with machines?

This course consists of 12 lectures that describe the study and creation of social robots. Learning the ingredients is one thing, combining them in practice is another. Therefore, lectures are flanked by practical exercises during the work groups, where you will learn how to formally model a theory but also how to draw on your own creativity to design a robot or a robot application. If possible (but this is dependent on the availability of participants), we will let your design be tested by real users. While working, we will have plenty of discussions about the societal impact of robots on work, privacy, security, ethical behavior, our self-image as humans, and the like.

Theoretical explorations will not be limited to communication science but will stretch to science philosophy (i.e. epistemics) and the theory of creativity and innovation as well.

Additional Information Teaching Methods

Lectures and work groups. Students will be trained to access their individual creativity and ingenuity.

Method of Assessment

1. Individual examination (‘tentamen’) (50% final grade)
2. Short graded assignments (group-based) (40% final grade)
3. Individual participation in the work groups (10% final grade)

Examination, each assignment, and participation each must be passed (‘voldoende’) to be able to pass the course.

Literature

Hyperlinks to scientific articles from different disciplines (humanities, communication, and computer science) will be made available on Canvas.

Additional Information Target Audience

- Elective for Master students Communication Science.
- Master students of Linguistics, Computer Science (i.e. AI and CHI), Health Science, Industrial Design Engineering (i.e. interaction design), and foreign exchange students are welcome.

Additional Information

- This course will be taught in English. We encourage students from different backgrounds to apply (Social Sciences, Psychology, Linguistics, Computer Science/ AI, industrial design, eHealth, education...).
- Programming skills or any kind of technical background are not required.

Recommended background knowledge

BA level cognitive theories and concepts (E.g. psychology, linguistics, communication, artificial intelligence, industrial design engineering, education, or health intervention).

BA level knowledge of experimental design, empirical testing and/or formal modelling.

Useful but not mandatory: structured questionnaire design, statistics, coding (any language), electronics, mechanics.

Socially Intelligent Robotics

Course Code	XM_0074
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. M.E.U. Ligthart
Examiner	dr. M.E.U. Ligthart
Teaching Staff	dr. M.E.U. Ligthart, prof. dr. K.V. Hindriks
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

The overall objective of this course is that students will be able to apply basic design skills to create interaction designs for a social robot and develop key social skills for the social robot using AI techniques.

Students will gain an understanding about social robotics and related AI techniques to control a social robot (e.g., conversational AI, computer vision aimed at interacting with people, expressing gestures, etc.) and apply this knowledge by designing a social human-robot interaction with the Nao robot in groups. (Applying Knowledge and Understanding)

Students will evaluate the social skills they developed for the robot with a pilot study (Making Judgments).

Students will present their design, prototype and evaluation results and communicate about their individual roles

within their group (Communication).

Students will be challenged to take the initiative and direct their own learning by identifying relevant social skills for the robot given a selected interaction context, creating a design rationale that is grounded in existing (multi-disciplinary) literature, and subsequently implementing that in a Nao robot (Learning Skills).

Course Content

In this course we will take a user-centered approach to the design of social robots and look into AI techniques for developing a social robot that can interact with human users. We will look at the basic cognitive skills we expect a social robot to have, including visual perception (e.g. face recognition), speech recognition and dialogue, emotional expression through body language, and the architecture for integrating these various skills to execute them on the robot.

Additional Information Teaching Methods

Lectures and practical group work. Approximately 7h / week mandatory practical session.

Method of Assessment

There will be a mid-term exam (20%) to assess the theoretical understanding of the concepts covered in the lectures and selected literature . This is an individual assessment.

Furthermore, you will be asked to complete a practical assignment with a group of students. Grades for each of the main deliverables will contribute to the final grade and will be weight as follows:

- Robot software (20%)
- Design document (50%)
- Final presentation (10%)

Every group member is expected to keep track of their weekly progress in the form of a logbook. In case we find clear differences in what and how much individual group members have contributed to the final result (deliverables), we may take this into account and differentiate grades for individual group members. We will also use input from the teaching assistants, who will discuss with your group each week to establish such differences.

A minimum grade of 5.5 for each component is required.

It will not be possible to redo the practical assignment (no resit).

Literature

A brief course manual will be made available. The main literature used will consist of existing literature (papers and other materials) on social robotics and related AI techniques.

Additional Information Target Audience

Master Artificial Intelligence

Additional Information

Important: Students who cannot attend the practical sessions on campus cannot join this course (because these practical sessions are crucial for the learning goals).

Custom Course Registration

The capacity for this course is 100 students.

Recommended background knowledge

Students should have the programming skills (Python) and ability to learn to use a programming framework for social robots that will be made available in the course.

Structural Bioinformatics

Course Code	X_405019
Credits	6

Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	R. Kumar
Examiner	dr. ir. K.A. Feenstra
Teaching Staff	dr. J.M. Urtasun Elizari, dr. H. Mouhib, dr. ir. K.A. Feenstra, R. Kumar
Teaching method(s)	Computer lab, Written partial exam, Lecture

Course Objective

Why Structural Bioinformatics?

The function of a protein is determined by its three-dimensional structure, and therefore structural information is crucial for understanding the working of proteins. However, experiments, prediction and simulation of protein structures remain difficult. This course will provide you with an overview of existing computational techniques, to validate, simulate, predict and analyse protein structures. More importantly, it will provide practical knowledge about how and when to use such techniques.

After completing the course successfully,

- You understand the basics of statistical thermodynamics and the relation to protein structures, evolution and function [Knowledge and understanding]
- You are able to evaluate protein structures with knowledge of their experimental source, bias and validation techniques [Applying knowledge and understanding]
- You are able to compare different protein structures, and evaluate similarity [Applying knowledge and understanding]
- You know how and when to use structure prediction methods [Making judgements; Applying knowledge and understanding]
- You can perform calculations on protein structures to gain new biological insights [Applying knowledge and understanding]
- You are able to compare different simulation techniques for biological macro-molecules, and be able to analyse the simulated data computationally [Applying knowledge and understanding]
- You are able to read and understand scientific papers in the field of Structural Bioinformatics [Communication; Lifelong learning skills]
- You are able to independently search for answers to open problems and ask a question about something that is unclear [Communication, Lifelong learning skills]

Course Content

Theory:

- Protein and DNA structure sources
- Experimental methods
- Structure validation
- Protein fold prediction (from homology modelling to ab initio prediction)
- Structural classification and structural alignment
- Protein folding and energetics
- Molecular Dynamics & Monte Carlo simulation
- Function from structure

Practical:

- Obtaining geometric features from PDB files
- Homology modelling with Modeller
- Protein interaction as a 'computational experiment' (simulation)

Additional Information Teaching Methods

- 13 lectures (2 two-hour lectures per week)
- Exercises during/in between lectures
- 12 computer practicals (2 two-hour sessions per week)
- Feedback (theoretical and practical) will be given during the computer practical sessions.

Method of Assessment

The final grade for this course will consist of 50% practical work and 50% theoretical assessment.

Practical Assignments: (50%)
(assessment of practical assignments will be adjusted based on ongoing developments in generative language models)

Theoretical: (50%)

- Oral, written or digital exam (depending on number of course students) to assess lecture topics.
- As part of the exam research papers on Structural Bioinformatics topics need to be analysed in detail.
- You will be prepared for your exam through exercises and paper discussions during the lectures, as well as through the practical assignments.

Assessment and grading details will be posted on Canvas.

Entry Requirements

Bachelor in any science discipline (including medicine), with an interest in molecular biology and applying algorithmic approaches to protein structures.
Experience with programming (preferably python).
Basic understanding of calculus and of organic chemistry.
When in doubt please contact the coordinator.

Literature

- course material on canvas.vu.nl
- book: Marketa Zvelebil and Jeremy O. Baum. Understanding Bioinformatics. Garland Science 2008 ISBN-10: 0-8153-4024-9
- book "Introduction to Structural Bioinformatics": The intro is published on ArXiv [https://arxiv.org/abs/1801.09442], and the chapters are available from the table of contents. The book will also be posted on canvas.

Additional Information Target Audience

M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Artificial Intelligence
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)
We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. We will provide some help in setting this up at the start of the course.
Lectures may be recorded; these recordings will only be used within the course, including possibly in subsequent years as background information.

Recommended background knowledge

Some experience with the linux command line is preferred.

Subjectivity Mining

Course Code	L_AAMPLIN018
Credits	6
Period	P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Humanities
Course Coordinator	dr. E. Maks
Examiner	dr. E. Maks
Teaching Staff	dr. I. Markov, dr. E. Maks
Teaching method(s)	Seminar

Course Objective

This course aims to help students to understand the latest developments in hate speech detection, especially when it comes to automatically identifying harmful content using modern technology and machine learning. We will explore some of the recurrent issues in the field, like identifying subtle or implicit hate speech, the challenges of deciding what actually counts as hate speech, and how definitions can vary across datasets. The goal is to give students both the technical skills and the critical perspective needed to tackle these real-world challenges.

To achieve this objective, students learn to:

- Define and explain the key concepts related to hate speech, including its social and linguistic dimensions.
- Describe the role and importance of annotated texts and linguistic resources in developing Automated Hate Speech Detection (AHSD) systems.
- Analyze how the definition and operationalization of hate speech affect annotation reliability and the effectiveness of AHSD systems.
- Develop an experimental setup using various computational approaches, such as traditional machine learning methods, transformer-based architectures, prompt engineering techniques with LLMs and ensemble strategies.
- Perform both quantitative analysis and qualitative error analyses on the results of AHSD systems.
- Be able to critically appraise literature on hate speech and its automated detection, and interpret their own methodology and findings in the context of this wider literature.
- Demonstrate awareness of emerging challenges in the field, including multimodal and meme-based hate speech detection and the role of counter-narratives in online discourse.

Course Content

In recent decades, the number of digital texts that have been produced and become publicly available has increased exponentially. As such, a vast volume of texts now exists that can -in principle- be ‘read’ (that is, processed, analysed and interpreted) by appropriately instructed and programmed computers. The purposes for which such text mining can be used are enormous, ranging form the development of voting aids, stock market prediction and opinion mining (e.g., opinions toward contentious issues such as vaccination). Many of these texts are subjective: they contain sentiments, opinions, stances, beliefs, speculations and emotions (also called private states). Automated analysis of these private states starts with identifying the intentions of the speaker or writer of the message and its associated functions and syntactic structures, not to mention the choice of vocabulary and associated connotations.

The topic of the current edition of the Subjectivity Mining course is online hate speech and its automated detection. With the growth of online content the spread of hate speech has expanded, too. Hate speech, offensive language, cyberbullying, etc. create important societal problems as they can cause serious harm to the victims. A strong desire to control this spread has stirred increased interest in the potential for automated hate speech detection (AHSD). The automated detection of hate speech in digital texts is, however, associated with significant challenges These challenges derive from different sources:

- There are many different types of hate speech, varying in terms of degree of harmfulness, target, immediate threat, etc. Relatedly, hate speech has been differently defined and operationalized.
- The context of the hate speech (e.g., Facebook, Twitter, blogs, etc).
- The possibilities for creating manually annotated gold standards, to be used for assessing performance and training of AHSD.
- The sort of method and settings used for AHSD.

The overall objective of the course is not only to introduce students to the theory and methods of subjectivity mining. We also aim at improving their understanding of the factors that affect detection performance, help them to develop their skills in critically appraising specific instances of subjectivity mining (in this case AHSD) and in designing and implementing strategies for automated identification, taking into account the various relevant contextual factors. In other words: tools always produce results; the challenge is to develop the ability to critically interpret such results. We believe that, more than the technicalities, this will help students to become life-long learners in this rapidly expanding field of computational linguistics.

Additional Information Teaching Methods

Lectures and working groups (2 * 1,5 hour per week)

Method of Assessment

Weekly assignments and a final course paper

Literature

To be announced

Additional Information Target Audience

2nd year's Students of the Research Master's Humanities, in particular Linguistics (Human Language Technology) and master-students of Artificial Intelligence and Computer science and master-students of Business Analytics

Recommended background knowledge

1. Programming Skills (Python); AND
2. at least one machine learning course AND
3. (L_AAMPALG016) Introduction to Human Language Technology (L_AAMPALG016) OR (Natural Language Processing Technology) L_AAMAALG005

The Social Web

Course Code	X_405086
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D. Ceolin
Examiner	dr. D. Ceolin
Teaching Staff	Z. Inayat, dr. D. Ceolin, dr. F. Ilievski
Teaching method(s)	Computer lab, Lecture

Course Objective

In this course, students will:

- Learn theory and methods concerning communication and interaction in a Web context (Knowledge and Understanding).
- Understand how user data is represented and distributed across devices in the context of the Social Web (Knowledge and Understanding).
- Collect and prepare Web data, apply the above skills in practical assignments (Apply Knowledge and Understanding)
- Reflect on social and ethical responsibilities linked to the application of these methods (Making Judgements).
- Communicate knowledge and judgements in a final paper (Communication).

Course Content

This course will cover theory, methods and techniques for:

- personalization for Web applications;
- Web user & context modeling;
- user-generated content and metadata;
- multi-device interaction;
- usage of social-web data.

The course takes a data science approach, using Jupyter notebooks to collect and analyse data, with a research paper describing the data science study as the main outcome of the course.

Additional Information Teaching Methods

Lectures (1x week)
Practical sessions (1x week)
Presentation seminars (1x week)

Method of Assessment

The weighted average of:

- 70% research paper (group assignment A),
- 10% group presentation (P),
- and 20% individual exam (E).

A minimum grade of 5.5 for each component is required.

Compensation: it is not possible to compensate one component with another.

Mode of re-examination: if insufficient, each component may be revised and resubmitted once.

Literature

- Course lecture slides;
- Selected articles, videos and Web links

All materials for each lecture will be provided through Canvas.

Additional Information Target Audience

Master Artificial Intelligence
Master Computer Science
Master Information Sciences
Master Computer Security

Additional Information

The course has a maximum enrolment limit of 120. The students of the Master Information Sciences, for whom it is compulsory, have priority over the students that take this course as an elective or a constrained choice course.

The students that take this course as an elective or a constrained choice course run the risk of not being able to participate, even if they have registered on time, if the maximum enrolment limit has been reached.

The students of the Master Information Sciences, for whom it is compulsory, run the risk of being initially refused if the enrolment limit has been reached at the time of registration. However this will be adjusted manually after registration has been closed so that all the students of the Master Information Sciences will be able to participate.

Recommended background knowledge

Basic Python programming skills.